

Sets, Combinatorics, and Probability

CHAPTER OBJECTIVES

After studying this chapter, you will be able to:

- Use the notation of set theory.
- Find the power set of a finite set.
- Find the union, intersection, difference, complement, and Cartesian product of sets.
- Identify binary and unary operations on a set.
- Prove set identities.
- Recognize that not all sets are countable.
- Apply the multiplication principle and the addition principle to solve counting problems.
- Use decision trees to solve counting problems.
- Use the principle of inclusion and exclusion to find the number of elements in the union of sets.
- Use the pigeonhole principle to decide when certain events must occur.
- Use the formulas for permutations and combinations of r objects, with and without repetition, from a set of n distinct objects.
- Find the number of distinct permutations of n objects that are not all distinct.
- Generate all permutations of n distinct objects, and all combinations of r out of n distinct objects.
- Find the probability of an event given that all outcomes are equally likely, that a probability distribution has been assigned to the outcomes, or that another event has already occurred.
- Compute the expected value of a quantity with an assigned probability distribution.
- Use the binomial theorem to expand $(a + b)^n$.

You survey the 87 computer users who subscribe to your electronic newsletter in preparation for the release of your new software product. The results of your survey reveal that 68 have a Windows-based system available to them, 34 have a Linux system available, and 30 have access to a Mac. In addition, 19 have access to both Windows and Linux systems, 11 have access to both Linux systems and Macs, and 23 can use both Macs and Windows.

Question: How many of your subscribers have access to all three types of systems?

This is an example of a counting problem; you want to count the number of elements in a certain collection or set—the set of all subscribers with access to all three systems. A formula that easily solves this counting problem is developed in Section 4.3.

Set theory is one of the cornerstones of mathematics. Many concepts in mathematics and computer science can be conveniently expressed in the language of sets. Operations can be performed on sets to generate new sets. Although most sets of interest to computer scientists are finite or countable, there are sets with so many members that they cannot be enumerated. Set theory is discussed in Section 4.1.

It is often of interest to count the number of elements in a finite set. This may not be a trivial task. Section 4.2 provides some ground rules for counting the number of elements in a set consisting of the outcomes of an event. Counting the elements in such a set can be made manageable by breaking the event down into a sequence of subevents or into disjoint subevents that have no outcomes in common. Some specialized counting principles appear in Section 4.3. Section 4.4 provides formulas for counting the number of ways to arrange objects in a set and to select objects from a set, as well as algorithms to generate all the possible arrangements or selections. Section 4.5 discusses the binomial theorem, an algebraic result that can also be viewed as a consequence of the counting formulas. Finally, Section 4.6 extends “counting” to the more general idea of probability.

SECTION 4.1 SETS

Definitions are important in any science because they contribute to precise communication. However, if we look up a word in the dictionary, the definition is expressed using other words, which are defined using still other words, and so on. Thus, we have to have a starting point for definitions where the meaning is taken to be understood; our starting point in this discussion will be the idea of a set, a term that we will not formally define. Instead, we will simply use the intuitive idea that a set is a collection of objects. Usually all of the objects in a set share some common property (aside from that of belonging to the same set!); any object that has the property is a member of the set, and any object that does not have the property is not a member. (This is consistent with our use of the word *set* in Section 3.1, where we talked about the set of propositional well-formed formulas, the set of all strings of symbols from a finite alphabet, and the set of identifiers in some programming language.)

Notation

We use capital letters to denote sets and the symbol $\in$ to denote membership in a set. Thus $a \in A$ means that object a is a member, or element, of set A , and $b \notin A$ means that object b is not an element of set A . Braces are used to indicate a set.

EXAMPLE 1 If $A = \{\text{violet, chartreuse, burnt umber}\}$, then $\text{chartreuse} \in A$ and $\text{magenta} \notin A$. •

No ordering is imposed on the elements in a set; therefore $\{\text{violet, chartreuse, burnt umber}\}$ is the same as $\{\text{chartreuse, burnt umber, violet}\}$. Also, each element of a set is listed only once; it is redundant to list it again.

Two sets are **equal** if they contain the same elements. (In a definition, “if” really means “if and only if”; thus two sets are equal if and only if they contain the same elements.) Using predicate logic notation,

$$A = B \text{ means } (\forall x)[(x \in A \rightarrow x \in B) \wedge (x \in B \rightarrow x \in A)]$$

In describing a particular set, we have to identify its elements. For a **finite set** (one with n elements for some nonnegative integer n), we might do this by simply listing all the elements, as in set A of Example 1. Although it is impossible to list all elements of an **infinite set** (one that is not finite), for some infinite sets we can indicate a pattern for listing elements indefinitely. Thus, we might write $\{2, 4, 6, \dots\}$ to express the set S of all positive even integers. (Although this is a common practice, the danger exists that the reader will not see the pattern that the writer has in mind.) S can also be defined recursively by giving an explicit member of S and then describing other members of S in terms of already known members. For example,

1. $2 \in S$
2. If $n \in S$, then $(n + 2) \in S$

But the clearest way to describe this particular set S is to describe the characterizing property of the set elements in words and write

$$S = \{x \mid x \text{ is a positive even integer}\}$$

read as “the set of all x such that x is a positive even integer.”

We’ve now given three ways to describe a set:

1. List (or partially list) its elements.
2. Use recursion to describe how to generate the set elements.
3. Describe a property P that characterizes the set elements.

Later in this section we’ll see that there are sets for which the first approach won’t work; often the second approach is difficult to use. The third method is usually the best choice.

The notation for a set S whose elements are characterized as having property P is $\{x \mid P(x)\}$. Property P here is a *unary predicate*; this term was introduced in Chapter 1. For any given x , $P(x)$ is either true or false. In fact, the formal logic notation of Chapter 1 again comes to the rescue to clarify what we mean by a characterizing property of a set’s elements:

$$S = \{x \mid P(x)\} \text{ means } (\forall x)[(x \in S \rightarrow P(x)) \wedge (P(x) \rightarrow x \in S)]$$

In words, every element of S has property P and everything that has property P is an element of S .

PRACTICE 1 Describe each of the following sets by listing its elements.

- a. $\{x \mid x \text{ is an integer and } 3 < x \leq 7\}$
- b. $\{x \mid x \text{ is a month with exactly 30 days}\}$
- c. $\{x \mid x \text{ is the capital of the United States}\}$

PRACTICE 2 Describe each of the following sets by giving a characterizing property.

- a. $\{1, 4, 9, 16\}$
- b. $\{\text{the butcher, the baker, the candlestick maker}\}$
- c. $\{2, 3, 5, 7, 11, 13, 17, \dots\}$

It is convenient to name certain standard sets so that we can refer to them easily. We will use

- $\mathbb{N}$ = set of all nonnegative integers (note that $0 \in \mathbb{N}$)
- $\mathbb{Z}$ = set of all integers
- $\mathbb{Q}$ = set of all rational numbers
- $\mathbb{R}$ = set of all real numbers
- $\mathbb{C}$ = set of all complex numbers

Sometimes we will also want to talk about the set with no elements (the **empty set**, or **null set**), denoted by $\emptyset$ or $\{ \}$. For example, if $S = \{x \mid x \in \mathbb{N} \text{ and } x < 0\}$, then $S = \emptyset$. Note that $\emptyset$, the set with no elements, is not the same as $\{\emptyset\}$, which is a set with a single element where the single element is the empty set.

EXAMPLE 2 Suppose that a set A is described as

$$A = \{x \mid (\exists y)(y \in \{0, 1, 2\} \text{ and } x = y^3)\}$$

Because y is not a free variable here, this is still of the form $A = \{x \mid P(x)\}$. The members of A can be found by letting y assume each of the values 0, 1, and 2 and then taking the cube of each such value. Therefore $A = \{0, 1, 8\}$. For

$$B = \{x \mid x \in \mathbb{N} \text{ and } (\exists y)(y \in \mathbb{N} \text{ and } x \leq y)\}$$

choosing $y = 0$ gives $x = 0$; choosing $y = 1$ gives $x = 0$ or 1; choosing $y = 2$ gives $x = 0, 1$, or 2; and so on. In other words, B consists of all nonnegative integers that are less than or equal to some nonnegative integer, which means that $B = \mathbb{N}$. For the set

$$C = \{x \mid x \in \mathbb{N} \text{ and } (\forall y)(y \in \mathbb{N} \rightarrow x \leq y)\}$$

0 is the only nonnegative integer that is less than or equal to every nonnegative integer, so $C = \{0\}$. ■

PRACTICE 3 Describe each set.

- a. $A = \{x \mid x \in \mathbb{N} \text{ and } (\forall y)(y \in \{2, 3, 4, 5\} \rightarrow x \geq y)\}$
 - b. $B = \{x \mid (\exists y)(\exists z)(y \in \{1, 2\} \text{ and } z \in \{2, 3\} \text{ and } x = y + z)\}$
-

Relationships Between Sets

For $A = \{2, 3, 5, 12\}$ and $B = \{2, 3, 4, 5, 9, 12\}$, every member of A is also a member of B . When this happens, A is said to be a *subset* of B .

PRACTICE 4 Complete the definition: A is a **subset** of B if

$$(\forall x)(x \in A \rightarrow \underline{\hspace{2cm}})$$
■

If A is a subset of B , we denote the relationship by $A \subseteq B$. If $A \subseteq B$ but $A \neq B$ (there is at least one element of B that is not an element of A), then A is a **proper subset** of B , denoted by $A \subset B$.

PRACTICE 5 Use formal logic notation to define $A \subset B$.

EXAMPLE 3 Let

$$\begin{aligned} A &= \{1, 7, 9, 15\} \\ B &= \{7, 9\} \\ C &= \{7, 9, 15, 20\} \end{aligned}$$

Then the following statements (among others) are all true:

$$\begin{aligned} B &\subseteq C & 15 &\in C \\ B &\subseteq A & \{7, 9\} &\subseteq B \\ B &\subset A & \{7\} &\subset A \\ A &\not\subseteq C & \emptyset &\subseteq C \end{aligned}$$

REMINDER

Be sure you understand the difference between the symbols $\in$ (element of) and $\subseteq$ (subset of).

The last statement ($\emptyset \subseteq C$) is true because the statement $(\forall x)(x \in \emptyset \rightarrow x \in C)$ is true because $x \in \emptyset$ is always false.

PRACTICE 6 Let

$$\begin{aligned} A &= \{x \mid x \in \mathbb{N} \text{ and } x \geq 5\} \\ B &= \{10, 12, 16, 20\} \\ C &= \{x \mid (\exists y)(y \in \mathbb{N} \text{ and } x = 2y)\} \end{aligned}$$

Which of the following statements are true?

- | | |
|---------------------------------|--|
| a. $B \subseteq C$ | g. $\{12\} \in B$ |
| b. $B \subset A$ | h. $\{12\} \subseteq B$ |
| c. $A \subseteq C$ | i. $\{x \mid x \in \mathbb{N} \text{ and } x < 20\} \not\subseteq B$ |
| d. $26 \in C$ | j. $5 \subseteq A$ |
| e. $\{11, 12, 13\} \subseteq A$ | k. $\{\emptyset\} \subseteq B$ |
| f. $\{11, 12, 13\} \subset C$ | l. $\emptyset \notin A$ |

Suppose that $B = \{x \mid P(x)\}$ and that $A \subseteq B$. Because every element of A is also an element of B , and P is a property characterizing all elements of B , then every element in A also has property $P(x)$. The elements of A “inherit” property P . In fact, to prove that $A \subseteq B$, we pick an arbitrary $x \in A$ and show that $P(x)$

holds. If A is a proper subset of B , A 's elements will usually have some additional characterizing property not shared by all elements of B . (This is the same notion of “inheritance” that prevails when a child type, or subtype, or derived type is defined in an object-oriented programming language. The child type inherits all of the properties and operations from the parent type with the addition of specialized local properties or operations as needed.)

EXAMPLE 4 Let

$$B = \{x \mid x \text{ is a multiple of } 4\}$$

and let

$$A = \{x \mid x \text{ is a multiple of } 8\}$$

Then we have $A \subseteq B$. To prove it, let $x \in A$; note that x is a completely arbitrary member of A . We must show that x satisfies the characterizing property of B ; in other words, we must show that x is a multiple of 4. Because we have $x \in A$, x satisfies the characterizing property of A ; that is, x is a multiple of 8 and thus we can write $x = m \cdot 8$ for some integer m . This equation can be written as $x = m \cdot 2 \cdot 4$ or $x = k \cdot 4$, where $k = 2m$, so k is an integer. This shows that x is a multiple of 4, and therefore $x \in B$.

There are numbers (like 12) that are multiples of 4 but not multiples of 8, so $A \subset B$. Another way to describe A is

$$A = \{x \mid x = k \cdot 4 \text{ and } k \text{ is an even number}\}$$

In this form it is clear that A 's elements have inherited the characterizing property of B —being a multiple of 4—but that there is an additional restriction that makes A less general than B . ●

PRACTICE 7 Let

$$A = \{x \mid x \in \mathbb{R} \text{ and } x^2 - 4x + 3 = 0\}$$

$$B = \{x \mid x \in \mathbb{N} \text{ and } 1 \leq x \leq 4\}$$

Prove that $A \subset B$. ■

We know that A and B are equal sets if they have the same elements. We can restate this equality in terms of subsets: $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$. Proving set inclusion in both directions is the usual way to establish the equality of two sets.

EXAMPLE 5

We will prove that $\{x \mid x \in \mathbb{N} \text{ and } x^2 < 15\} = \{x \mid x \in \mathbb{N} \text{ and } 2x < 7\}$.

Let $A = \{x \mid x \in \mathbb{N} \text{ and } x^2 < 15\}$ and $B = \{x \mid x \in \mathbb{N} \text{ and } 2x < 7\}$. To show that $A = B$, we show $A \subseteq B$ and $B \subseteq A$. For $A \subseteq B$, we must choose an arbitrary member of A —that is, anything satisfying the characterizing property of A —and show that it also satisfies the characterizing property of B . Let $x \in A$. Then x is a nonnegative integer satisfying the inequality $x^2 < 15$. The nonnegative integers with squares less than 15 are 0, 1, 2, and 3, so these integers are the members of A . The double of each of these nonnegative integers is a number less than 7. Hence, each member of A is a member of B , and $A \subseteq B$.

Now we show $B \subseteq A$. Any member of B is a nonnegative integer whose double is less than 7. These numbers are 0, 1, 2, and 3, each of which has a square less than 15, so $B \subseteq A$. ●

Sets of Sets

For a set S , we can form a new set whose elements are all of the subsets of S . This new set is called the **powerset** of S , $\wp(S)$.

EXAMPLE 6

For $S = \{0, 1\}$, $\wp(S) = \{\emptyset, \{0\}, \{1\}, \{0, 1\}\}$. Note that the members of the power set of a set are themselves sets. ●

REMINDER

To find $\wp(S)$, start with $\emptyset$.
Then add sets taking 1 element from S at a time, then 2 elements at a time, then 3 at a time, and so forth.

For any set S , $\wp(S)$ will always have at least $\emptyset$ and S itself as members, since $\emptyset \subseteq S$ and $S \subseteq S$ are always true.

PRACTICE 8 For $A = \{1, 2, 3\}$, what is $\wp(A)$? ■

In Practice 8, A has 3 elements and $\wp(A)$ has 8 elements. Try finding $\wp(S)$ for other sets S until you can guess the answer to the following practice problem.

PRACTICE 9 If S has n elements, then $\wp(S)$ has _____ elements. (Does your answer work for $n = 0$, too?) ■

There are several ways we can show that for a set S with n elements, $\wp(S)$ will have 2^n elements. The following proof uses induction. For the basis step of the induction, we let $n = 0$. The only set with 0 elements is $\emptyset$. The only subset of $\emptyset$ is $\emptyset$, so $\wp(\emptyset) = \{\emptyset\}$, a set with $1 = 2^0$ elements. We assume that for any set with k elements, the power set has 2^k elements.

Now let S have $k + 1$ elements and put one of these elements, call it x , aside. The remaining set has k elements, so by our inductive assumption, its power set has 2^k elements. Each of these elements is also a member of $\wp(S)$. The only members of $\wp(S)$ not counted by this procedure are those including element x . All the subsets including x can be found by taking all those subsets not including x (of which there are 2^k) and throwing in the x ; thus, there will be 2^k subsets including x . Altogether, there are 2^k subsets without x and 2^k subsets with x , or $2^k + 2^k = 2 \cdot 2^k = 2^{k+1}$ subsets. Therefore, $\wp(S)$ has 2^{k+1} elements.

Analogy with the truth tables of Section 1.1 is another way to show that $\wp(S)$ has 2^n elements for a set S with n elements. There we had n statement letters and showed that there were 2^n true-false combinations among these letters. But we can also think of each true-false combination as representing a particular subset, with T indicating membership and F indicating nonmembership in that subset. (For example, the row of the truth table with all statement letters F corresponds to the empty set.) Thus, the number of true-false combinations among n statement letters equals the number of subsets of a set with n elements; both are 2^n .

Binary and Unary Operations

By itself a set is not very interesting until we do something with its elements. For example, we can perform several arithmetic operations on elements of the set $\mathbb{Z}$. We might subtract two integers, or we might take the negative of an integer. Subtraction acts on two integers; it is a *binary* operation on $\mathbb{Z}$. Negation acts on one integer; it is a *unary* operation on $\mathbb{Z}$.

To see exactly what is involved in a binary operation, let's look at subtraction more closely. For any two integers x and y , $x - y$ produces an answer and only one answer, and that answer is always an integer. Finally, subtraction is performed on an **ordered pair** of numbers. For example, $7 - 5$ does not produce the same result as $5 - 7$. An ordered pair is denoted by (x, y) , where x is the first component of the ordered pair and y is the second component. Order is important in an ordered pair; thus, the sets $\{1, 2\}$ and $\{2, 1\}$ are equal, but the ordered pairs $(1, 2)$ and $(2, 1)$ are not. You are probably familiar with ordered pairs used as coordinates to locate a point in the plane. The point $(1, 2)$ is different from the point $(2, 1)$. Two ordered pairs (x, y) and (u, v) are equal only when it is the case that $x = u$ and $y = v$.

PRACTICE 10 Given that $(2x - y, x + y) = (7, -1)$, solve for x and y .

PRACTICE 11 Let $S = \{3, 4\}$. List all the ordered pairs (x, y) of elements of S .

We will generalize the properties of subtraction on the integers to define a binary operation $\circ$ on a set S . The symbol $\circ$ is merely a placeholder; in any specific discussion, it will be replaced by the appropriate operation symbol, such as a subtraction sign.

• **DEFINITION BINARY OPERATION**

$\circ$ is a **binary operation** on a set S if for every ordered pair (x, y) of elements of S , $x \circ y$ exists, is unique, and is a member of S .

In other words, if $\circ$ is a binary operation on S , then for any two values x and y in S , $x \circ y$ produces one and only one answer, and that answer belongs to S . That the value $x \circ y$ always exists and is unique. It is described by saying that the binary operation $\circ$ is **well-defined**. The property that $x \circ y$ always belongs to S is described by saying that S is **closed** under the operation $\circ$. Uniqueness does not mean that the result of a binary operation occurs only once; it means that for a given x and y , there is only one result. For subtraction, there are many x and y values such that $x - y = 7$, but for a given x and y , like $x = 5$ and $y = 2$, there is only one answer for $x - y$.

EXAMPLE 7 Addition, subtraction, and multiplication are all binary operations on $\mathbb{Z}$. For example, when we perform addition on the ordered pair of integers (x, y) , $x + y$ exists and is a unique integer. •

EXAMPLE 8 The logical operations of conjunction, disjunction, implication, and equivalence are binary operations on the set of propositional wffs. If P and Q are propositional wffs, then $P \wedge Q$, $P \vee Q$, $P \rightarrow Q$, and $P \leftrightarrow Q$ are unique propositional wffs. •

A candidate $\circ$ for an operation can fail to be a binary operation on a set S in any of three ways: (1) There are elements $x, y \in S$ for which $x \circ y$ does not exist; (2) there are elements $x, y \in S$ for which $x \circ y$ gives more than one result; or (3) there are elements $x, y \in S$ for which $x \circ y$ does not belong to S .

EXAMPLE 9 Division is not a binary operation on $\mathbb{Z}$ because $x \div 0$ does not exist. •

EXAMPLE 10 Define $x \circ y$ on $\mathbb{N}$ by

$$x \circ y = \begin{cases} 1 & \text{if } x \geq 5 \\ 0 & \text{if } x < 5 \end{cases}$$

Then, by the first part of the definition for $\circ$, $5 \circ 1 = 1$, but by its second part, $5 \circ 1 = 0$. Thus, $\circ$ is not well-defined on $\mathbb{N}$ because the result of $5 \circ 1$ is not unique. •

EXAMPLE 11 Subtraction is not a binary operation on $\mathbb{N}$ because $\mathbb{N}$ is not closed under subtraction. (For example, $1 - 10 \notin \mathbb{N}$.) •

For $\#$ to be a **unary operation** on a set S , it must be true that for any $x \in S$, $x\#$ is well-defined and S is closed under $\#$; in other words, for any $x \in S$, $x\#$ exists, is unique, and is a member of S . We do not have a unary operation if any of these conditions is not met.

EXAMPLE 12

Let $x\#$ be defined by $x\# = -x$ so that $x\#$ is the negative of x . Then $\#$ is a unary operation on $\mathbb{Z}$ but not on $\mathbb{N}$ because $\mathbb{N}$ is not closed under $\#$. ●

EXAMPLE 13

The logical connective of negation is a unary operation on the set of propositional wffs. If P is a propositional wff, then P' is a unique propositional wff. ●

From these examples it is clear that whether $\circ$ (or $\#$) is a binary (or unary) operation can depend not only on its definition but also on the set involved.

PRACTICE 12

Which of the following candidates are neither binary nor unary operations on the given sets? Why not?

- a. $x \circ y = x \div y$; $S =$ set of all positive integers
- b. $x \circ y = x \div y$; $S =$ set of all positive rational numbers
- c. $x \circ y = x^y$; $S = \mathbb{R}$
- d. $x \circ y =$ maximum of x and y ; $S = \mathbb{N}$
- e. $x^\# = \sqrt{x}$; $S =$ set of all positive real numbers
- f. $x^\# =$ solution to equation $(x^\#)^2 = x$; $S = \mathbb{C}$

So far, all our binary operations have been defined by means of a description or an equation. Suppose S is a finite set, $S = \{x_1, x_2, \dots, x_n\}$. Then a binary operation $\circ$ on S can be defined by an $n \times n$ table, where element i, j (i th row and j th column) denotes $x_i \circ x_j$.

EXAMPLE 14

Let $S = \{2, 5, 9\}$, and let $\circ$ be defined by the table

$\circ$	2	5	9
2	2	2	9
5	5	9	2
9	5	5	9

Thus, $2 \circ 5 = 2$ and $9 \circ 2 = 5$. Inspecting the table, we see that $\circ$ is a binary operation on S . ●

Operations on Sets

Most of the operations we have seen operate on numbers, but we can also operate on sets. Given an arbitrary set S , we can define some binary and unary

operations on the set $\wp(S)$. S in this case is called the **universal set** or the **universe of discourse**. The universal set defines the context of the objects being discussed. If $S = \mathbb{Z}$, for example, then all subsets will contain only integers.

A binary operation on $\wp(S)$ must act on any two subsets of S to produce a unique subset of S . There are at least two natural ways in which this can happen.

EXAMPLE 15

Let S be the set of all students at Silicon U. Then the members of $\wp(S)$ are sets of students. Let A be the set of computer science majors, and let B be the set of business majors. Both A and B belong to $\wp(S)$. A new set of students can be defined that consists of everybody who is majoring in either computer science or business (or both); this set is called the *union* of A and B . Another new set can be defined that consists of everybody who is majoring in both computer science and business. This set (which might be empty) is called the *intersection* of A and B .

• **DEFINITION UNION AND INTERSECTION OF SETS**

Let $A, B \in \wp(S)$. The **union** of A and B , denoted by $A \cup B$, is $\{x \mid x \in A \text{ or } x \in B\}$. The **intersection** of A and B , denoted by $A \cap B$, is $\{x \mid x \in A \text{ and } x \in B\}$.

EXAMPLE 16

Let $A = \{1, 3, 5, 7, 9\}$ and $B = \{3, 5, 6, 10, 11\}$. Here we may consider A and B as members of $\wp(\mathbb{N})$. Then $A \cup B = \{1, 3, 5, 6, 7, 9, 10, 11\}$ and $A \cap B = \{3, 5\}$. Both $A \cup B$ and $A \cap B$ are members of $\wp(\mathbb{N})$.

PRACTICE 13

Let $A, B \in \wp(S)$ for any set S . Is it always the case that $A \cap B \subseteq A \cup B$?

We can use *Venn diagrams* (named for the nineteenth-century British mathematician John Venn) to visualize the binary operations of union and intersection. The shaded areas in Figures 4.1 and 4.2 illustrate the set that results from performing the binary operation on the two given sets.

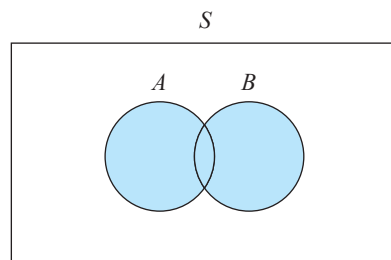


Figure 4.1 $A \cup B$

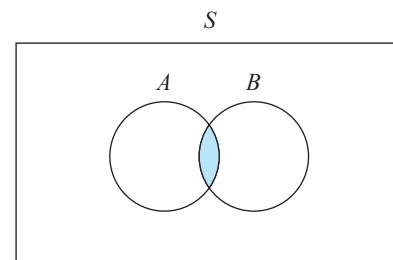


Figure 4.2 $A \cap B$

We will define one unary operation on $\wp(S)$.

● **DEFINITION** **COMPLEMENT OF A SET**

For a set $A \in \wp(S)$, the **complement** of A , A' , is $\{x \mid x \in S \text{ and } x \notin A\}$.

PRACTICE 14 Illustrate A' in a Venn diagram. ■

EXAMPLE 17

In Section 1.1 we discussed the use of logical connectives to formulate Web search queries. A query such as

“used cars” AND (Ford OR Buick) AND NOT trucks

is asking the search engine to return a set of pages (or more properly, a set of links to pages). If

U = set of used car pages

F = set of Ford pages

B = set of Buick pages

T = set of truck pages

then

$$U \cap (F \cup B) \cap T'$$

represents the set of Web pages that is the desired result of the query. ●

Another binary operation on sets A and B in $\wp(S)$ is **set difference**: $A - B = \{x \mid x \in A \text{ and } x \notin B\}$. This operation can be rewritten as $A - B = \{x \mid x \in A \text{ and } x \in B'\}$ and, finally, as $A - B = A \cap B'$.

PRACTICE 15 Illustrate $A - B$ in a Venn diagram. ■

Two sets A and B such that $A \cap B = \emptyset$ are said to be **disjoint**. Thus, $A - B$ and $B - A$, for example, are disjoint sets.

EXAMPLE 18

Let

$$A = \{x \mid x \text{ is an even nonnegative integer}\}$$

$$B = \{x \mid (\exists y)(y \in \mathbb{N} \text{ and } x = 2y + 1)\}$$

$$C = \{x \mid (\exists y)(y \in \mathbb{N} \text{ and } x = 4y)\}$$

be subsets of $\mathbb{N}$. Because B represents the set of nonnegative odd integers, A and B are disjoint sets. Also, every nonnegative integer is either even or odd, so $A \cup B = \mathbb{N}$. These two facts also tell us that $A' = B$. Every multiple of 4 is an even number, so C is a subset of A , from which it follows that $A \cup C = A$. C is in fact a proper subset of A , and $A - C = \{x \mid (\exists y)(y \in \mathbb{N} \text{ and } x = 4y + 2)\}$. ●

PRACTICE 16 Let

$$A = \{1, 2, 3, 5, 10\}$$

$$B = \{2, 4, 7, 8, 9\}$$

$$C = \{5, 8, 10\}$$

be subsets of $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Find

- a. $A \cup B$ b. $A - C$ c. $B' \cap (A \cup C)$

We will define one final operation using elements of $\wp(S)$.

● **DEFINITION** **CARTESIAN PRODUCT**

Let A and B be subsets of S . The **Cartesian product (cross product)** of A and B , denoted by $A \times B$, is defined by

$$A \times B = \{(x, y) \mid x \in A \text{ and } y \in B\}$$

Thus, the Cartesian product of two sets A and B is the set of all ordered pairs whose first component comes from A and whose second component comes from B . The cross product is not a binary operation on $\wp(S)$. Although it acts on an ordered pair of members of $\wp(S)$ and gives a unique result, the resulting set is not, in general, a subset of S . The elements are not members of S but ordered pairs of members of S . So the resulting set is not a member of $\wp(S)$. The closure property for a binary operation fails to hold.

Because we will often be interested in the cross product of a set with itself, we will abbreviate $A \times A$ as A^2 ; in general, we use A^n to mean the set of all ordered n -tuples $(x_1, x_2, \dots, x_n)$ of elements of A .

PRACTICE 17 Let $A = \{1, 2\}$ and $B = \{3, 4\}$.

- a. Find $A \times B$. b. Find $B \times A$. c. Find A^2 . d. Find A^3 .

Set Identities

There are many set equalities involving the operations of union, intersection, difference, and complementation that are true for all subsets of a given set S . Because they are independent of the particular subsets used, these equalities are called set identities. Some basic set identities follow. The names and forms of these identities are very similar to the tautological equivalences of Section 1.1 (check back and compare). We will see in Chapter 8 that this similarity is not a coincidence.

BASIC SET IDENTITIES

1a. $A \cup B = B \cup A$	1b. $A \cap B = B \cap A$	(commutative properties)
2a. $(A \cup B) \cup C = A \cup (B \cup C)$	2b. $(A \cap B) \cap C = A \cap (B \cap C)$	(associative properties)
3a. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	3b. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	(distributive properties)
4a. $A \cup \emptyset = A$	4b. $A \cap S = A$	(identity properties)
5a. $A \cup A' = S$	5b. $A \cap A' = \emptyset$	(complement properties)

(Note that 2a allows us to write $A \cup B \cup C$ with no need for parentheses; 2b allows us to write $A \cap B \cap C$.)

EXAMPLE 19

Let's prove identity 3a. We might draw Venn diagrams for each side of the equation and see that they look the same. However, identity 3a is supposed to hold for all subsets A , B , and C , and whatever one picture we draw cannot be completely general. Thus, if we draw A and B disjoint, that's a special case, but if we draw A and B not disjoint, that doesn't take care of the case where A and B are disjoint. To do a proof by Venn diagrams requires a picture for each possible case, and the more sets involved (A , B , and C in this problem), the more cases there are. To avoid drawing a picture for each case, let's prove set equality by proving set inclusion in each direction. Thus, we want to prove

$$A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$$

and also

$$(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$$

To show that $A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$, we let x be an arbitrary member of $A \cup (B \cap C)$. Then we can proceed as follows:

$$\begin{aligned}
 x \in A \cup (B \cap C) &\rightarrow x \in A \text{ or } x \in (B \cap C) \\
 &\rightarrow x \in A \text{ or } (x \in B \text{ and } x \in C) \\
 &\rightarrow (x \in A \text{ or } x \in B) \text{ and } (x \in A \text{ or } x \in C) \\
 &\rightarrow x \in (A \cup B) \text{ and } x \in (A \cup C) \\
 &\rightarrow x \in (A \cup B) \cap (A \cup C)
 \end{aligned}$$

To show that $(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$, we reverse the above argument. ●

PRACTICE 18 Prove identity 4a.

Once we have proved the set identities in this list, we can use them to prove other set identities. Just as the tautological equivalences of propositional logic represent recipes or patterns for transforming wffs, the set identities represent patterns for transforming set expressions. And, as with tautologies, the set identity can be applied only when the set expression exactly matches the pattern.

EXAMPLE 20

We can use the basic set identities to prove

$$[A \cup (B \cap C)] \cap ([A' \cup (B \cap C)] \cap (B \cap C)') = \emptyset$$

for A , B , and C any subsets of S . In the following proof, the number to the right is that of the basic set identity used to validate each step. The first step uses identity 2b because the expression

$$[A \cup (B \cap C)] \cap ([A' \cup (B \cap C)] \cap (B \cap C)')$$

matches the right side of 2b, $A \cap (B \cap C)$ where A is $[A \cup (B \cap C)]$, B is $[A' \cup (B \cap C)]$, and C is $(B \cap C)'$

$$\begin{aligned} & [A \cup (B \cap C)] \cap ([A' \cup (B \cap C)] \cap (B \cap C)') \\ &= ([A \cup (B \cap C)] \cap [A' \cup (B \cap C)]) \cap (B \cap C)' & (2b) \\ &= [(B \cap C) \cup A] \cap [(B \cap C) \cup A'] \cap (B \cap C)' & (1a \text{ twice}) \\ &= [(B \cap C) \cup (A \cap A')] \cap (B \cap C)' & (3a) \\ &= [(B \cap C) \cup \emptyset] \cap (B \cap C)' & (5b) \\ &= (B \cap C) \cap (B \cap C)' & (4a) \\ &= \emptyset & (5b) \end{aligned}$$

REMINDER

You must match the pattern of a set identity in order to use it. In the set identities, A , B , and C can represent any sets.

The **dual** for each set identity in our list also appears in the list. The dual is obtained by interchanging $\cup$ and $\cap$ and interchanging S and $\emptyset$. The dual of the identity in Example 20 is

$$[A \cap (B \cup C)] \cup ([A' \cap (B \cup C)] \cup (B \cup C)') = S$$

which we could prove true by replacing each basic set identity used in the proof of Example 20 with its dual. Because this method always works, any time we have proved a set identity by using the basic identities, we have also proved its dual.

PRACTICE 19

- a. Using the basic set identities, establish the set identity

$$[C \cap (A \cup B)] \cup [(A \cup B) \cap C'] = A \cup B$$

(A , B , and C are any subsets of S .)

- b. State the dual identity that you now know is true.

Table 4.1 summarizes the approaches to proving set identities.

TABLE 4.1	
Method	Comment
Draw a Venn diagram	Not a good plan because no one diagram fits all cases and it will not prove the general identity.
Establish set inclusion in each direction	Take an arbitrary member of one side and show it belongs to the other side, and conversely.
Use already proved identities	Be sure to match the pattern of the identity you want to use.

Countable and Uncountable Sets

In a finite set S , we can always designate one element as the first member, s_1 , another element as the second member, s_2 , and so forth. If there are k elements in the set, then these can be listed in the order we have selected:

$$s_1, s_2, \dots, s_k$$

This list represents the entire set S . The number of elements in a finite set is the **cardinality** of the set, so this would be a set of cardinality k , denoted by $|S| = k$.

If the set is infinite, we may still be able to select a first element s_1 , a second element s_2 , and so forth, so that the list

$$s_1, s_2, s_3, \dots$$

represents all elements of the set. Every element of the set will eventually appear in this list. Such an infinite set is said to be **denumerable**. Both finite and denumerable sets are **countable** sets because we can count, or enumerate, all of their elements. Being countable does not mean that we can state the total number of elements in the set; rather, it means that we can say, “Here is a first one,” “Here is a second one,” and so on, through the set. There are, however, infinite sets that are **uncountable**. In an uncountable set, the set is so big that there is no way to count out the elements and get the whole set in the process. Before we prove that uncountable sets exist, let’s look at some denumerable (countably infinite) sets.

EXAMPLE 21

The set $\mathbb{N}$ is denumerable.

To prove denumerability, we need only exhibit a counting scheme. For the set $\mathbb{N}$ of nonnegative integers, it is clear that

$$0, 1, 2, 3, \dots$$

is an enumeration that will eventually include every member of the set. ●

PRACTICE 20

Prove that the set of even positive integers is denumerable. ■

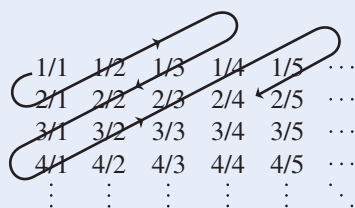
EXAMPLE 22

The set $\mathbb{Q}^+$ of positive rational numbers is denumerable.

We assume that each positive rational number is written as a fraction of positive integers. We can write all such fractions having the numerator 1 in one row, all those having the numerator 2 in a second row, and so on:

$$\begin{array}{cccccc} 1/1 & 1/2 & 1/3 & 1/4 & 1/5 & \cdots \\ 2/1 & 2/2 & 2/3 & 2/4 & 2/5 & \cdots \\ 3/1 & 3/2 & 3/3 & 3/4 & 3/5 & \cdots \\ 4/1 & 4/2 & 4/3 & 4/4 & 4/5 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{array}$$

To show that the set of all fractions in this array is denumerable, we will thread an arrow through the entire array, beginning with $1/1$; following the arrow gives an enumeration of the set. Thus the fraction $1/3$ is the fourth member in this enumeration:



Therefore the set represented by the array is denumerable. Note that our path through the array must “spread out” from one corner. If we begin to follow just the first row or just the first column, for example, we will never finish it to get on to other rows (or columns).

To obtain an enumeration of $\mathbb{Q}^+$, we use the enumeration of the set shown but eliminate any fractions not in lowest terms. This avoids the problem of listing both $1/2$ and $2/4$, for example, which represent the same positive rational. The enumeration of $\mathbb{Q}^+$ thus begins with

$$1/1, 2/1, 1/2, 1/3, 3/1, 4/1, \dots$$

For example, we have eliminated $2/2$, which reduces to $1/1$.

PRACTICE 21

What is the 11th fraction in the above enumeration? What is the 11th positive rational? ■

Now let's show that there is an uncountable (not countable) infinite set. The proof technique that seems appropriate to prove that set A does *not* have property B is to assume that A does have property B and look for a contradiction. The proof in Example 23 is a very famous proof by contradiction known as **Cantor's diagonalization method**, after Georg Cantor, the nineteenth-century German mathematician known as the “father of set theory.”

EXAMPLE 23

We will show that the set of all the real numbers between 0 and 1 is uncountable.

We will write such numbers in decimal form; thus any member of the set can be written as

$$0.d_1d_2d_3 \dots$$

Now let us assume that our set is countable. Therefore some enumeration of the set exists. A number such as $0.24999999 \dots$ can be written in alternative form as $0.2500000 \dots$ (see Exercise 102 for an explanation of why these are alternative representations of the same number). To avoid writing the same element twice in our enumeration, we will choose (arbitrarily) to always use the former representation and not the latter. We can depict an enumeration of the set as follows, where d_{ij} is the j th decimal digit in the i th number in the enumeration:

$$\begin{array}{r} 0.d_{11}d_{12}d_{13} \dots \\ 0.d_{21}d_{22}d_{23} \dots \\ 0.d_{31}d_{32}d_{33} \dots \\ \vdots \quad \vdots \quad \vdots \end{array}$$

We now construct a real number $p = 0.p_1p_2p_3 \dots$ as follows: p_i is always chosen to be 5 if $d_{ii} \neq 5$ and 6 if $d_{ii} = 5$. Thus p is a real number between 0 and 1. For instance, if the enumeration begins with

$$\begin{array}{l} 0.342134 \dots \\ 0.257001 \dots \\ 0.546122 \dots \\ 0.716525 \dots \end{array}$$

then $d_{11} = 3$, $d_{22} = 5$, $d_{33} = 6$, and $d_{44} = 5$, so $p_1 = 5$, $p_2 = 6$, $p_3 = 5$, and $p_4 = 6$. Thus p begins with $0.5656 \dots$

If we compare p with the enumeration of the set, p differs from the first number at the first decimal digit, from the second number at the second decimal digit, from the third number at the third decimal digit, and so on.

$$\begin{array}{l} 0.\textcircled{3}42134\dots \\ 0.2\textcircled{5}7001\dots \\ 0.54\textcircled{6}122\dots \\ 0.716\textcircled{5}25\dots \\ \vdots \end{array}$$

Therefore p does not agree with any of the representations in the enumeration. Furthermore, because p contains no 0s to the right of the decimal, it is not the alternative representation of any of the numbers in the enumeration. Therefore p is a real number between 0 and 1 different from any other number in the enumeration, yet the enumeration was supposed to include all members of the set. Here, then, is the contradiction, and the set of all real numbers between 0 and 1 is indeed uncountable. (You can see why this proof is called a “diagonalization method.”) ●

Although it is interesting and perhaps surprising to learn that there are uncountable sets, we are usually concerned with countable sets. A computer, of course, can manage only finite sets. In the rest of this chapter, we too, limit our attention to finite sets and various ways to count their elements.

SECTION 4.1 REVIEW

TECHNIQUES

- Describe sets by a list of elements and by a characterizing property.
- Prove that one set is a subset of another.
- Find the power set of a set.
- Check that the required properties for a binary or unary operation are satisfied.
- **W** Form new sets by taking the union, intersection, complement, and cross product of sets.
- **W** Prove set identities by showing set inclusion in each direction or using the basic set identities.
- Demonstrate the denumerability of certain sets.
- Use the Cantor diagonalization method to prove that certain sets are uncountable.

MAIN IDEAS

- Sets are unordered collections of objects that can be related (equal sets, subsets, etc.) or combined (union, intersection, etc.).
- Certain standard sets have their own notation.
- The power set of a set with n elements has 2^n elements.
- Basic set identities exist (in dual pairs) and can be used to prove other set identities; once an identity is proved in this manner, its dual is also true.
- Countable sets can be enumerated, and uncountable sets exist.

EXERCISES 4.1

- Let $S = \{2, 5, 17, 27\}$. Which of the following expressions are true?
 - $5 \in S$
 - $2 + 5 \in S$
 - $\emptyset \in S$
 - $S \in S$
- Let $B = \{x \mid x \in \mathbb{Q} \text{ and } -1 < x < 2\}$. Which of the following expressions are true?
 - $0 \in B$
 - $-1 \in B$
 - $-0.84 \in B$
 - $\sqrt{2} \in B$
- How many different sets are described here? What are they?

$\{2, 3, 4\}$	$\emptyset$
$\{x \mid x \text{ is the first letter of cat, bat, or apple}\}$	$\{x \mid x \text{ is the first letter of cat, bat, and apple}\}$
$\{x \mid x \in \mathbb{N} \text{ and } 2 \leq x \leq 4\}$	$\{2, a, 3, b, 4, c\}$
$\{a, b, c\}$	$\{3, 4, 2\}$
- How many different sets are described here? What are they?
 - $\{x \mid x = F(n) \wedge n \in \{5, 6, 7\}\}$ [$F(n)$ is a Fibonacci number]
 - $\{x \mid x \mid 24\}$ [x divides 24]
 - $\{1, 2, 3, 4\}$
 - $\{5, 8, 13\}$
 - $\{x \mid x \in \mathbb{N} \wedge 0 < x \leq 4\}$
 - $\{x \mid x \in \varphi(5)\}$ [$\varphi(n)$ is the Euler phi function]
 - $\{12, 2, 6, 24, 8, 3, 1, 4\}$
 - $\{x \mid x \text{ is a digit in the decimal equivalent of the Roman numeral MCCXXXIV}\}$

5. Describe each of the following sets by listing its elements:
- $\{x \mid x \in \mathbb{N} \text{ and } x^2 < 25\}$
 - $\{x \mid x \in \mathbb{N} \text{ and } x \text{ is even and } 2 < x < 11\}$
 - $\{x \mid x \text{ is one of the first three U.S. presidents}\}$
 - $\{x \mid x \in \mathbb{R} \text{ and } x^2 = -1\}$
 - $\{x \mid x \text{ is one of the New England states}\}$
 - $\{x \mid x \in \mathbb{Z} \text{ and } |x| < 4\}$ ($|x|$ denotes the absolute value function)
6. Describe each of the following sets by listing its elements:
- $\{x \mid x \in \mathbb{N} \text{ and } x^2 - 5x + 6 = 0\}$
 - $\{x \mid x \in \mathbb{R} \text{ and } x^2 = 7\}$
 - $\{x \mid x \in \mathbb{N} \text{ and } x^2 - 2x - 8 = 0\}$
7. Describe each of the following sets by giving a characterizing property:
- $\{1, 2, 3, 4, 5\}$
 - $\{1, 3, 5, 7, 9, 11, \dots\}$
 - $\{\text{Melchior, Gaspar, Balthazar}\}$
 - $\{0, 1, 10, 11, 100, 101, 110, 111, 1000, \dots\}$
8. Describe each of the following sets:
- $\{x \mid x \in \mathbb{N} \text{ and } (\exists q)(q \in \{2, 3\} \text{ and } x = 2q)\}$
 - $\{x \mid x \in \mathbb{N} \text{ and } (\exists y)(\exists z)(y \in \{0, 1\} \text{ and } z \in \{3, 4\} \text{ and } y < x < z)\}$
 - $\{x \mid x \in \mathbb{N} \text{ and } (\forall y)(y \text{ even} \rightarrow x \neq y)\}$
9. Given the description of a set A as $A = \{2, 4, 8, \dots\}$, do you think $16 \in A$?
10. What is the cardinality of each of the following sets?
- $S = \{a, \{a, \{a\}\}\}$
 - $S = \{\{a\}, \{\{a\}\}\}$
 - $S = \{\emptyset\}$
 - $S = \{a, \{\emptyset\}, \emptyset\}$
 - $S = \{\emptyset, \{\emptyset, \{\emptyset\}\}, \{\emptyset, \{\emptyset, \{\emptyset\}\}\}\}$
11. Let

$$A = \{2, 5, 7\}$$

$$B = \{1, 2, 4, 7, 8\}$$

$$C = \{7, 8\}$$

Which of the following statements are true?

- $5 \subseteq A$
 - $C \subseteq B$
 - $\emptyset \in A$
 - $7 \in B$
 - $\{2, 5\} \subseteq A$
 - $\emptyset \subseteq C$
12. Let

$$A = \{x \mid x \in \mathbb{N} \text{ and } 1 < x < 50\}$$

$$B = \{x \mid x \in \mathbb{R} \text{ and } 1 < x < 50\}$$

$$C = \{x \mid x \in \mathbb{Z} \text{ and } |x| \geq 25\}$$

Which of the following statements are true?

- a. $A \subseteq B$ e. $\sqrt{3} \in B$
 b. $17 \in A$ f. $\{0, 1, 2\} \subseteq A$
 c. $A \subseteq C$ g. $\emptyset \in B$
 d. $-40 \in C$ h. $\{x \mid x \in \mathbb{Z} \text{ and } x^2 > 625\} \subseteq C$

13. Let

$$\begin{aligned} R &= \{1, 3, \pi, 4.1, 9, 10\} & T &= \{1, 3, \pi\} \\ S &= \{\{1\}, 3, 9, 10\} & U &= \{\{1, 3, \pi\}, 1\} \end{aligned}$$

Which of the following statements are true? For those that are not, why not?

- a. $S \subseteq R$ e. $\{1\} \subseteq T$
 b. $1 \in R$ f. $\{1\} \subseteq S$
 c. $1 \in S$ g. $T \subset R$
 d. $1 \subseteq U$

14. Let

$$\begin{aligned} R &= \{1, 3, \pi, 4.1, 9, 10\} & T &= \{1, 3, \pi\} \\ S &= \{\{1\}, 3, 9, 10\} & U &= \{\{1, 3, \pi\}, 1\} \end{aligned}$$

Which of the following statements are true? For those that are not, why not?

- a. $\{1\} \in S$ e. $T \notin R$
 b. $\emptyset \subseteq S$ f. $T \subseteq R$
 c. $T \subseteq U$ g. $S \subseteq \{1, 3, 9, 10\}$
 d. $T \in U$

15. Let

$$A = \{a, \{a\}, \{\{a\}\}\} \quad B = \{a\} \quad C = \{\emptyset, \{a, \{a\}\}\}$$

Which of the following statements are true? For those that are not, where do they fail?

- a. $B \subseteq A$ f. $\{a, \{a\}\} \in A$
 b. $B \in A$ g. $\{a, \{a\}\} \subseteq A$
 c. $C \subseteq A$ h. $B \subseteq C$
 d. $\emptyset \subseteq C$ i. $\{\{a\}\} \subseteq A$
 e. $\emptyset \in C$

16. Let

$$A = \{\emptyset, \{\emptyset, \{\emptyset\}\}\} \quad B = \emptyset \quad C = \{\emptyset\} \quad D = \{\emptyset, \{\emptyset\}\}$$

Which of the following statements are true? For those that are not, where do they fail?

- a. $C \subseteq A$ f. $C = B$
 b. $C \in A$ g. $C \subseteq D$
 c. $B \in A$ h. $C \in D$
 d. $B \subseteq A$ i. $D \subseteq A$
 e. $B \in C$

17. Let

$$A = \{(x, y) \mid (x, y) \text{ lies within 3 units of the point } (1, 4)\}$$

and

$$B = \{(x, y) \mid (x - 1)^2 + (y - 4)^2 \leq 25\}$$

Prove that $A \subset B$.

18. Let

$$A = \{x \mid x \in \mathbb{R} \text{ and } x^2 - 4x + 3 < 0\}$$

and

$$B = \{x \mid x \in \mathbb{R} \text{ and } 0 < x < 6\}$$

Prove that $A \subset B$.

19. Program QUAD finds and prints solutions to quadratic equations of the form $ax^2 + bx + c = 0$. Program EVEN lists all the even integers from $-2n$ to $2n$. Let Q denote the set of values output by QUAD and E denote the set of values output by EVEN.
- Show that for $a = 1$, $b = -2$, $c = -24$, and $n = 50$, $Q \subseteq E$.
 - Show that for the same values of a , b , and c , but a value for n of 2, $Q \not\subseteq E$.
20. Let $A = \{x \mid \cos(x/2) = 0\}$ and $B = \{x \mid \sin x = 0\}$. Prove that $A \subseteq B$.
21. Which of the following statements are true for all sets A , B , and C ?
- If $A \subseteq B$ and $B \subseteq A$, then $A = B$.
 - $\{\emptyset\} = \emptyset$
 - $\{\emptyset\} = \{0\}$
 - $\emptyset \in \{\emptyset\}$
 - $\emptyset \subseteq A$
22. Which of the following statements are true for all sets A , B , and C ?
- $\emptyset \in A$
 - $\{\emptyset\} = \{\{\emptyset\}\}$
 - If $A \subset B$ and $B \subseteq C$, then $A \subset C$.
 - If $A \neq B$ and $B \neq C$, then $A \neq C$.
 - If $A \in B$ and $B \notin C$, then $A \notin C$.
23. Prove that if $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.
24. Prove that if $A' \subseteq B'$, then $B \subseteq A$.
25. Prove that for any integer $n \geq 2$, a set with n elements has $n(n - 1)/2$ subsets that contain exactly two elements.
26. Prove that for any integer $n \geq 3$, a set with n elements has $n(n - 1)(n - 2)/6$ subsets that contain exactly three elements. (*Hint*: Use Exercise 25.)
27. Find $\wp(S)$ for $S = \{a\}$.
28. Find $\wp(S)$ for $S = \{a, b\}$.
29. Find $\wp(S)$ for $S = \{1, 2, 3, 4\}$. How many elements do you expect this set to have?
30. Find $\wp(S)$ for $S = \{\emptyset\}$.

31. Find $\wp(S)$ for $S = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}$.
32. Find $\wp(\wp(S))$ for $S = \{a, b\}$.
33. What can be said about A if $\wp(A) = \{\emptyset, \{x\}, \{y\}, \{x, y\}\}$?
34. What can be said about A if $\wp(A) = \{\emptyset, \{a\}, \{\{a\}\}\}$?
35. Prove that if $\wp(A) = \wp(B)$, then $A = B$.
36. Prove that if $A \subseteq B$, then $\wp(A) \subseteq \wp(B)$.
37. Solve for x and y .
- a. $(y, x + 2) = (5, 3)$ b. $(2x, y) = (16, 7)$ c. $(2x - y, x + y) = (-2, 5)$
38. a. Recall that ordered pairs must have the property that $(x, y) = (u, v)$ if and only if $x = u$ and $y = v$. Prove that $\{\{x\}, \{x, y\}\} = \{\{u\}, \{u, v\}\}$ if and only if $x = u$ and $y = v$. Therefore, although we know that $(x, y) \neq \{x, y\}$, we can define the ordered pair (x, y) as the set $\{\{x\}, \{x, y\}\}$.
- b. Show by an example that we cannot define the ordered triple (x, y, z) as the set $\{\{x\}, \{x, y\}, \{x, y, z\}\}$.
39. Which of the following candidates are binary or unary operations on the given sets? For those that are not, where do they fail?
- a. $x \circ y = x + 1$; $S = \mathbb{N}$
- b. $x \circ y = x + y - 1$; $S = \mathbb{N}$
- c. $x \circ y = \begin{cases} x - 1 & \text{if } x \text{ is odd} \\ x & \text{if } x \text{ is even} \end{cases}$ $S = \mathbb{Z}$
- d. $x\# = \ln x$; $S = \mathbb{R}$
40. Which of the following candidates are binary or unary operations on the given sets? For those that are not, where do they fail?
- a. $x\# = x^2$; $S = \mathbb{Z}$
- b.
- | | | | |
|---------|---|---|---|
| $\circ$ | 1 | 2 | 3 |
| 1 | 1 | 2 | 3 |
| 2 | 2 | 3 | 4 |
| 3 | 3 | 4 | 5 |
- $S = \{1, 2, 3\}$
- c. $x \circ y =$ that fraction, x or y , with the smaller denominator; $S =$ set of all fractions.
- d. $x \circ y =$ that person, x or y , whose name appears first in an alphabetical sort; $S =$ set of 10 people with different names.
41. Which of the following candidates are binary or unary operations on the given sets? For those that are not, where do they fail?
- a. $x \circ y = \begin{cases} 1/x & \text{if } x \text{ is positive} \\ 1/(-x) & \text{if } x \text{ is negative} \end{cases}$ $S = \mathbb{R}$
- b. $x \circ y = xy$ (concatenation); $S =$ set of all finite-length strings of symbols from the set $\{p, q, r\}$
- c. $x\# = \lfloor x \rfloor$ where $\lfloor x \rfloor$ denotes the greatest integer less than or equal to x ; $S = \mathbb{R}$
- d. $x \circ y = \min(x, y)$; $S = \mathbb{N}$
42. Which of the following candidates are binary or unary operations on the given sets? For those that are not, where do they fail?
- a. $x \circ y =$ greatest common multiple of x and y ; $S = \mathbb{N}$
- b. $x \circ y = x + y$; $S =$ the set of Fibonacci numbers

- c. $x\# =$ the string that is the reverse of x ; $S =$ set of all finite-length strings of symbols from the set $\{p, q, r\}$
 d. $x \circ y = x + y$; $S = \mathbb{R} - \mathbb{Q}$
43. How many different unary operations can be defined on a set with n elements? (*Hint*: Think about filling in a table.)
44. How many different binary operations can be defined on a set with n elements? (*Hint*: Think about filling in a table.)
45. We have written binary operations in *infix* notation, where the operation symbol appears between the two operands, as in $A + B$. Evaluation of a complicated arithmetic expression is more efficient when the operations are written in *postfix* notation, where the operation symbol appears after the two operands, as in $AB+$. Many compilers change expressions in a computer program from infix to postfix form. One way to produce an equivalent postfix expression from an infix expression is to write the infix expression with a full set of parentheses, move each operator to replace its corresponding right parenthesis, and then eliminate all left parentheses. (Parentheses are not required in postfix notation.) Thus,

$$A * B + C$$

becomes, when fully parenthesized,

$$((A * B) + C)$$

and the postfix notation is $AB * C+$. Rewrite each of the following expressions in postfix notation:

- a. $(A + B) * (C - D)$
 b. $A ** B - C * D$ (**denotes exponentiation)
 c. $A * C + B / (C + D * B)$
46. Evaluate the following postfix expressions (see Exercise 45):
 a. $2\ 4 * 5 +$ b. $5\ 1 + 2 / 1 -$ c. $3\ 4 + 5\ 1 - *$
47. Let

$$\begin{aligned} A &= \{p, q, r, s\} \\ B &= \{r, t, v\} \\ C &= \{p, s, t, u\} \end{aligned}$$

be subsets of $S = \{p, q, r, s, t, u, v, w\}$. Find

- a. $B \cap C$ b. $A \cup C$ c. C' d. $A \cap B \cap C$
48. Let

$$\begin{aligned} A &= \{p, q, r, s\} \\ B &= \{r, t, v\} \\ C &= \{p, s, t, u\} \end{aligned}$$

be subsets of $S = \{p, q, r, s, t, u, v, w\}$. Find

- a. $B - C$ b. $(A \cup B)'$ c. $A \times B$ d. $(A \cup B) \cap C'$

49. Let

$$A = \{2, 4, 5, 6, 8\}$$

$$B = \{1, 4, 5, 9\}$$

$$C = \{x | x \in \mathbb{Z} \text{ and } 2 \leq x < 5\}$$

be subsets of $S = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Find

a. $A \cup B$ e. $A - B$

b. $A \cap B$ f. A'

c. $A \cap C$ g. $A \cap A'$

d. $B \cup C$

50. Let

$$A = \{2, 4, 5, 6, 8\}$$

$$B = \{1, 4, 5, 9\}$$

$$C = \{x | x \in \mathbb{Z} \text{ and } 2 \leq x < 5\}$$

be subsets of $S = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Find

a. $(A \cap B)'$ e. $(C' \cup B)'$

b. $C - B$ f. $B \times C$

c. $(C \cap B) \cup A'$ g. $(B - A) \cup C$

d. $(B - A)' \cap (A - B)$

51. Let

$$A = \{a, \{a\}, \{\{a\}\}\}$$

$$B = \{\emptyset, \{a\}, \{a, \{a\}\}\}$$

$$C = \{a\}$$

be subsets of $S = \{\emptyset, a, \{a\}, \{\{a\}\}, \{a, \{a\}\}\}$. Find

a. $A \cap C$ d. $\emptyset \cap B$ f. $A' \cap B$

b. $B \cap C'$ e. $(B \cup C) \cap A$ g. $\{\emptyset\} \cap B$

c. $A \cup B$

52. Let

$$A = \{x | x \text{ is the name of a former president of the United States}\}$$

$$B = \{\text{Adams, Hamilton, Jefferson, Grant}\}$$

$$C = \{x | x \text{ is the name of a state}\}$$

Find

a. $A \cap B$ b. $A \cap C$ c. $B \cap C$

53. Let $S = A \times B$ where $A = \{2, 3, 4\}$ and $B = \{3, 5\}$. Which of the following statements are true?

- a. $A \subseteq S$ d. $(5, 4) \in S$
 b. $3 \in S$ e. $\emptyset \subseteq S$
 c. $(3, 3) \in S$ f. $\{(2, 5)\} \subseteq S$

54. Let

$$\begin{aligned} A &= \{x \mid x \text{ is a word that appears before } \textit{dog} \text{ in an English language dictionary}\} \\ B &= \{x \mid x \text{ is a word that appears after } \textit{canary} \text{ in an English language dictionary}\} \\ C &= \{x \mid x \text{ is a word of more than four letters}\} \end{aligned}$$

Which of the following statements are true?

- a. $B \subseteq C$
 b. $A \cup B = \{x \mid x \text{ is a word in an English language dictionary}\}$
 c. $\textit{cat} \in B \cap C'$
 d. $\textit{bamboo} \in A - B$

55. Consider the following subsets of $\mathbb{Z}$:

$$\begin{aligned} A &= \{x \mid (\exists y)(y \in \mathbb{Z} \text{ and } y \geq 4 \text{ and } x = 3y)\} \\ B &= \{x \mid (\exists y)(y \in \mathbb{Z} \text{ and } x = 2y)\} \\ C &= \{x \mid x \in \mathbb{Z} \text{ and } |x| \leq 10\} \end{aligned}$$

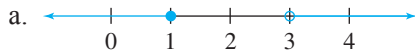
Using set operations, describe each of the following sets in terms of A , B , and C .

- a. set of all odd integers
 b. $\{-10, -8, -6, -4, -2, 0, 2, 4, 6, 8, 10\}$
 c. $\{x \mid (\exists y)(y \in \mathbb{Z} \text{ and } y \geq 2 \text{ and } x = 6y)\}$
 d. $\{-9, -7, -5, -3, -1, 1, 3, 5, 7, 9\}$
 e. $\{x \mid (\exists y)(y \in \mathbb{Z} \text{ and } y \geq 5 \text{ and } x = 2y + 1)\} \cup \{x \mid (\exists y)(y \in \mathbb{Z} \text{ and } y \leq -5 \text{ and } x = 2y - 1)\}$

56. Let

$$\begin{aligned} A &= \{x \mid x \in \mathbb{R} \text{ and } 1 < x \leq 3\} \\ B &= \{x \mid x \in \mathbb{R} \text{ and } 2 \leq x \leq 5\} \end{aligned}$$

Using set operations, describe each of the sets shown in terms of A and B .



57. Consider the following subsets of the set of all students:

A = set of all computer science majors

B = set of all physics majors

C = set of all science majors

D = set of all female students

Using set operations, describe each of the following sets in terms of A , B , C , and D :

- set of all students not majoring in science
- set of all female physics majors
- set of all students majoring in both computer science and physics
- set of all male computer science majors

58. Consider the following subsets of the set of all students:

A = set of all computer science majors

B = set of all physics majors

C = set of all science majors

D = set of all female students

Using set operations, describe each of the following sets in terms of A , B , C , and D :

- set of all male students who are not physics majors
 - set of all science majors who are not computer science majors
 - set of all students who are female or science majors
 - set of all students who are science majors but are neither computer science nor physics majors
- Write the set expression for the desired results of the Web search query for sites pertaining to dogs that are not retrievers (Exercise 31, Section 1.1). Assume that D = set of dog pages, R = set of retriever pages.
 - Write the set expression for the desired results of the Web search query for sites pertaining to oil paintings by Van Gogh or Rembrandt but not Vermeer (Exercise 32, Section 1.1). Assume that O = set of oil painting pages, G = set of Van Gogh pages, R = set of Rembrandt pages, V = set of Vermeer pages.
 - Write the set expression for the desired results of the Web search query for sites pertaining to novels or plays about AIDS (Exercise 33, Section 1.1). Assume that N = set of novel pages, P = set of play pages, A = set of AIDS pages.
 - Write the set expression for the desired results of the Web search query for sites pertaining to coastal wetlands in Louisiana but not in Alabama (Exercise 34, Section 1.1). Assume that C = set of coastal wetlands pages, L = set of Louisiana pages, A = set of Alabama pages.
 - Which of the following statements are true for all sets A , B , and C ?
 - $A \cup A = A$
 - $B \cap B = B$
 - $(A \cap B)' = A' \cap B'$
 - $(A')' = A$
 - $A - B = (B - A)'$
 - $(A - B) \cap (B - A) = \emptyset$
 - Which of the following statements are true for all sets A , B , and C ?
 - If $A \cap B = \emptyset$, then $A \subset B$.
 - $B \times A = A \times B$
 - $\emptyset \times A = \emptyset$
 - $\emptyset \cap \{\emptyset\} = \emptyset$
 - $(A - B) \cup (B - C) = A - C$
 - $(A - C) \cap (A - B) = A - (B \cup C)$

65. Which of the following statements are true for all sets A , B , and C ?
- $A \cup (B \times C) = (A \cup B) \times (A \cup C)$
 - $A \times (B \cap C) = (A \times B) \cap (A \times C)$
 - $A \times \emptyset = \emptyset$
66. Which of the following statements are true for all sets A , B , and C ?
- $\wp(A) \times \wp(A) = \wp(A^2)$
 - $A \times (B \times C) = (A \times B) \times C$
 - $\wp(A \times B) = \wp(A) \times \wp(B)$
67. For each of the following statements, find general conditions on sets A and B to make the statement true:
- $A \cup B = A$
 - $A \cap B = A$
 - $A \cup \emptyset = \emptyset$
 - $B - A = \emptyset$
 - $A \cup B \subseteq A \cap B$
 - $A \times B = B \times A$
68. For any finite set S , $|S|$ denotes the number of elements in S . If $|A| = 3$ and $|B| = 4$, find
- $|A \times B|$
 - $|A^2|$
 - $|B^2|$
 - the maximum possible value for $|A \cap B|$
 - the minimum possible value for $|A \cup B|$
69. Prove that $(A \cap B) \subseteq A$ where A and B are arbitrary sets.
70. Prove that $A \subseteq (A \cup B)$ where A and B are arbitrary sets.
71. Prove that $\wp(A) \cap \wp(B) = \wp(A \cap B)$ where A and B are arbitrary sets.
72. Prove that $\wp(A) \cup \wp(B) \subseteq \wp(A \cup B)$ where A and B are arbitrary sets.
73. Prove that if $A \cup B = A - B$, then $B = \emptyset$. (*Hint*: Do a proof by contradiction.)
74. Prove that if $(A - B) \cup (B - A) = A \cup B$, then $A \cap B = \emptyset$. (*Hint*: Do a proof by contradiction.)
75. Prove that if $C \subseteq B - A$, then $A \cap C = \emptyset$.
76. Prove that if $(A - B) \cup B = A$, then $B \subseteq A$.
77. Prove that $A \subseteq B$ if and only if $A \cap B' = \emptyset$.
78. Prove that $(A \cap B) \cup C = A \cap (B \cup C)$ if and only if $C \subseteq A$.

Exercises 79 and 80 refer to a binary operation on sets called the *symmetric difference*, which is defined by $A \oplus B = (A - B) \cup (B - A)$.

79. a. Draw a Venn diagram to illustrate $A \oplus B$.
 b. For $A = \{3, 5, 7, 9\}$ and $B = \{2, 3, 4, 5, 6\}$, what is $A \oplus B$?
 c. Prove that $A \oplus B = (A \cup B) - (A \cap B)$ for arbitrary sets A and B .
80. a. For an arbitrary set A , what is $A \oplus A$? What is $\emptyset \oplus A$?
 b. Prove that $A \oplus B = B \oplus A$ for arbitrary sets A and B .
 c. For any sets A , B , and C , prove that $(A \oplus B) \oplus C = A \oplus (B \oplus C)$.
81. Verify the basic set identities on page 234 by showing set inclusion in each direction. (We have already done 3a and 4a.)
82. A and B are subsets of a set S . Prove the following set identities by showing set inclusion in each direction.
- $(A \cup B)' = A' \cap B'$
 - $(A \cap B)' = A' \cup B'$
- } De Morgan's laws

- c. $A \cup (B \cap A) = A$
 d. $(A \cap B')' \cup B = A' \cup B$
 e. $(A \cap B) \cup (A \cap B') = A$
 f. $[A \cap (B \cup C)]' = A' \cup (B' \cap C')$
83. A , B , and C are subsets of a set S . Prove the following set identities using the basic set identities listed in this section. Give a reason for each step. State the dual of each of these identities.
- a. $(A \cup B) \cap (A \cup B') = A$
 b. $[(A \cap C) \cap B] \cup [(A \cap C) \cap B'] \cup (A \cap C)' = S$
 c. $(A \cup C) \cap [(A \cap B) \cup (C' \cap B)] = A \cap B$
84. A is a subset of a set S . Prove the following set identities:
- a. $A \cup A = A$ d. $A \cup S = S$
 b. $A \cap A = A$ e. $(A')' = A$
 c. $A \cap \emptyset = \emptyset$
85. A , B , and C are subsets of a set S . Prove the following set identities by using previously proved identities, including those in Exercises 82–84. Give a reason for each step.
- a. $A \cap (B \cup A') = B \cap A$
 b. $(A \cup B) - C = (A - C) \cup (B - C)$
 c. $(A - B) - C = (A - C) - B$
86. A , B , and C are subsets of a set S . Prove the following set identities by using previously proved identities, including those in Exercises 82–84. Give a reason for each step.
- a. $[(A' \cup B') \cap A']' = A$
 b. $(A - B) - C = (A - C) - (B - C)$
 c. $A - (A - B) = A \cap B$
 d. $(A \cup B) - (A \cap B) = (A - B) \cup (B - A)$
87. The operation of set union can be defined as an n -ary operation for any integer $n \geq 2$.
- a. Give a definition similar to that for the union of two sets for $A_1 \cup A_2 \cup \cdots \cup A_n$.
 b. Give a recursive definition for $A_1 \cup A_2 \cup \cdots \cup A_n$.
88. Using the recursive definition of set union from Exercise 87(b), prove the generalized associative property of set union, which is that for any n with $n \geq 3$ and any p with $1 \leq p \leq n - 1$,
- $$(A_1 \cup A_2 \cup \cdots \cup A_p) \cup (A_{p+1} \cup A_{p+2} \cup \cdots \cup A_n) = A_1 \cup A_2 \cup \cdots \cup A_n$$
89. The operation of set intersection can be defined as an n -ary operation for any integer $n \geq 2$.
- a. Give a definition similar to that for the intersection of two sets for $A_1 \cap A_2 \cap \cdots \cap A_n$.
 b. Give a recursive definition for $A_1 \cap A_2 \cap \cdots \cap A_n$.
90. Using the recursive definition of set intersection from Exercise 89(b), prove the generalized associative property of set intersection, which is that for any n with $n \geq 3$ and any p with $1 \leq p \leq n - 1$,

$$(A_1 \cap A_2 \cap \cdots \cap A_p) \cap (A_{p+1} \cap A_{p+2} \cap \cdots \cap A_n) = A_1 \cap A_2 \cap \cdots \cap A_n$$

91. Prove that for subsets $A_1, A_2, \dots, A_n$ and B of a set S , the following generalized distributive properties hold, where $n \geq 2$. (See Exercises 87 and 89.)
- $B \cup (A_1 \cap A_2 \cap \dots \cap A_n) = (B \cup A_1) \cap (B \cup A_2) \cap \dots \cap (B \cup A_n)$
 - $B \cap (A_1 \cup A_2 \cup \dots \cup A_n) = (B \cap A_1) \cup (B \cap A_2) \cup \dots \cup (B \cap A_n)$
92. Prove that for subsets $A_1, A_2, \dots, A_n$ of a set S , the following generalized De Morgan's laws hold, where $n \geq 2$. (See Exercises 82, 87, and 89.)
- $(A_1 \cup A_2 \cup \dots \cup A_n)' = A_1' \cap A_2' \cap \dots \cap A_n'$
 - $(A_1 \cap A_2 \cap \dots \cap A_n)' = A_1' \cup A_2' \cup \dots \cup A_n'$
93. The operations of set union and set intersection can be extended to apply to an infinite family of sets. We may describe the family as the collection of all sets A_i , where i takes on any of the values of a fixed set I . Here, I is called the *index set* for the family. The union of the family, $\bigcup_{i \in I} A_i$, is defined by

$$\bigcup_{i \in I} A_i = \{x \mid x \text{ is a member of some } A_i\}$$

The intersection of the family, $\bigcap_{i \in I} A_i$, is defined by

$$\bigcap_{i \in I} A_i = \{x \mid x \text{ is a member of each } A_i\}.$$

- Let $I = \{1, 2, 3, \dots\}$, and for each $i \in I$, let A_i be the set of real numbers in the interval $(-1/i, 1/i)$. What is $\bigcup_{i \in I} A_i$? What is $\bigcap_{i \in I} A_i$?
 - Let $I = \{1, 2, 3, \dots\}$, and for each $i \in I$, let A_i be the set of real numbers in the interval $[-1/i, 1/i]$. What is $\bigcup_{i \in I} A_i$? What is $\bigcap_{i \in I} A_i$?
94. According to our use of the word “set,” if C is a subset of the universal set S , then every element of S either does or does not belong to C . In other words, the probability of a member x of S being a member of C is either 1 (x is a member of C) or 0 (x is not a member of C). C is a *fuzzy set* if every $x \in S$ has a probability p , $0 \leq p \leq 1$, of being a member of C . The probability p associated with x is an estimate of the likelihood that x may belong to C when the actual composition of C is unknown. Set operations can be done on fuzzy sets as follows: If element x has probability p_1 of membership in C and probability p_2 of membership in D , then the probability of x being a member of $C \cup D$, $C \cap D$, and C' is, respectively, $\max(p_1, p_2)$, $\min(p_1, p_2)$, and $1 - p_1$. (If we consider the statements $x \in C$ and $x \in D$ as propositional wffs A and B , respectively, with certain probabilistic truth values, then the probability of $x \in C \cup D$ is the probability that $A \vee B$ is true. The rules for fuzzy set operations then parallel the rules for *fuzzy logic*, discussed in Exercise 54, Section 1.1.)
- Let S be a set of possible disease-causing agents, $S = \{\text{genetics, virus, nutrition, bacteria, environment}\}$. The fuzzy sets AIDS and ALZHEIMERS are defined as AIDS = {genetics, 0.2; virus, 0.8; nutrition, 0.1; bacteria, 0.4; environment, 0.3} and ALZHEIMERS = {genetics, 0.7; virus, 0.4; nutrition, 0.3; bacteria, 0.3; environment, 0.4}.
- Find the fuzzy set AIDS $\cup$ ALZHEIMERS.
 - Find the fuzzy set AIDS $\cap$ ALZHEIMERS.
 - Find the fuzzy set (AIDS)'.

Exercises 95 and 96 complete the proof, begun in Section 2.2, that the second principle of induction, the first principle of induction, and the principle of well-ordering are all equivalent.

95. The *principle of well-ordering* says that every nonempty set of positive integers has a smallest member. Prove that the first principle of mathematical induction, that is,

$$\left. \begin{array}{l} 1. P(1) \text{ is true} \\ 2. (\forall k)[P(k) \text{ true} \rightarrow P(k+1) \text{ true}] \end{array} \right\} \rightarrow P(n) \text{ true for all positive integers } n$$

implies the principle of well-ordering. (*Hint:* Assume that the first principle of mathematical induction is valid, and use proof by contradiction to show that the principle of well-ordering is valid. Let T be a nonempty subset of the positive integers that has no smallest member. Let $P(n)$ be the property that every member of T is greater than n .)

96. Prove that the principle of well-ordering (see Exercise 95) implies the second principle of mathematical induction. *Hint:* Assume that the principle of well-ordering is valid, and let P be a property for which

- 1.' $P(1)$ is true
- 2.' $(\forall k)[P(r) \text{ true for all } r, 1 \leq r \leq k \rightarrow P(k + 1) \text{ true}]$

Let T be the subset of the positive integers defined by

$$T = \{t \mid P(t) \text{ is not true}\}$$

Show that T is the empty set.

97. Prove that the set of odd positive integers is denumerable.
98. Prove that the set $\mathbb{Z}$ of all integers is denumerable.
99. Prove that the set of all finite-length strings of the letter a is denumerable.
100. Prove that the set of all finite-length binary strings is denumerable.
101. Prove that the set $\mathbb{Z} \times \mathbb{Z}$ is denumerable.
102. In Example 23, the claim was made that $0.249999999 \dots$ is the same number as $0.250000000 \dots$. The first representation is a nonterminating decimal, and a calculus-type argument can be made that “in the limit” these are the same values. Here is a slightly different argument:
 - a. Let $n = 0.249999 \dots$.
 Compute $100n$ by multiplying both sides of this equation by 100.
 Subtract n from $100n$ to give a value for $99n$.
 Solve the resulting equation for n .
 - b. Let $m = 0.250000 \dots$.
 Compute $100m$ by multiplying both sides of this equation by 100.
 Subtract m from $100m$ to give a value for $99m$.
 Solve the resulting equation for m .
 - c. Compare the values of n and m .
103. Use Cantor’s diagonalization method to show that the set of all infinite sequences of positive integers is not countable.
104. Use Cantor’s diagonalization method to show that the set of all infinite strings of the letters $\{a, b\}$ is not countable.
105. Explain why the union of any two denumerable sets is denumerable.
106. Explain why any subset of a countable set is countable.
107. Sets can have sets as elements (see Exercise 13, for example). Let B be the set defined as follows:

$$B = \{S \mid S \text{ is a set and } S \notin S\}$$

Argue that both $B \in B$ and $B \notin B$ are true. This contradiction is called *Russell’s paradox*, after the famous philosopher and mathematician Bertrand Russell, who stated it in 1901. (A carefully constructed axiomatization of set theory puts some restrictions on what can be called a set. All ordinary sets are still sets, but peculiar sets that get us into trouble, like B in this exercise, seem to be avoided.)

SECTION 4.2 COUNTING

Combinatorics is the branch of mathematics that deals with counting. Counting questions are important whenever we have finite resources (How much storage does a particular database consume? How many users can a given computer configuration support?) or whenever we are interested in efficiency (How many computations does a particular algorithm involve?).

Counting questions often boils down to finding how many members there are in some finite set, that is, what is the cardinality of the set. This seemingly trivial question can be difficult to answer. We have already answered some “how many” questions—How many rows are there in a truth table with n statement letters, and how many subsets are there in a set with n elements? (Actually, as we’ve noted, these questions can be thought of as the same question.)

Multiplication Principle

We solved the truth table question by drawing a tree of possibilities. This tree suggests a general principle that can be used to solve many counting problems. Before we state the general principle, we look at another tree example.

EXAMPLE 24

A child is allowed to choose one jellybean out of two jellybeans, one red and one black, and one gummy bear out of three gummy bears, yellow, green, and white. How many different sets of candy can the child have?

We can solve this problem by breaking the task of choosing candy into two sequential tasks of choosing the jellybean and then choosing the gummy bear. The tree of Figure 4.3 shows that there are $2 \times 3 = 6$ possible outcomes: $\{R, Y\}$, $\{R, G\}$, $\{R, W\}$, $\{B, Y\}$, $\{B, G\}$, and $\{B, W\}$.

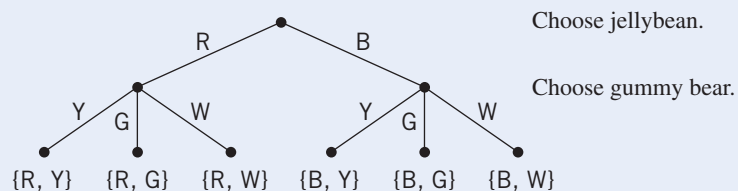


Figure 4.3

In this problem the sequence of events could be reversed; the child could choose the gummy bear first and the jellybean second, resulting in the tree of Figure 4.4, but the number of outcomes is the same ($3 \times 2 = 6$). Thinking of a sequence of successive events helps us solve the problem, but the sequencing is not a part of the problem since the set $\{R, Y\}$ is the same as the set $\{Y, R\}$.

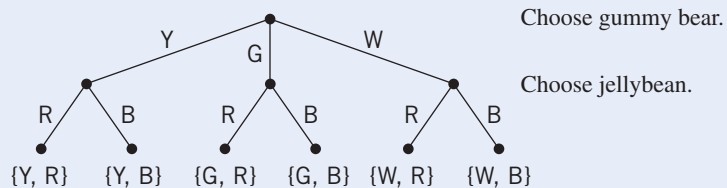


Figure 4.4

Both these trees are “balanced” in the sense that the second level has a fixed number of outcomes regardless of the outcome at the previous level. ●

Example 24 illustrates that the total number of outcomes for a sequence of events can be obtained by multiplying the number of outcomes for the first event by the number of outcomes for the second. This idea is summarized in the *multiplication principle*.

● **PRINCIPLE** **MULTIPLICATION PRINCIPLE**

If there are n_1 possible outcomes for a first event and n_2 possible outcomes for a second event, there are $n_1 \times n_2$ possible outcomes for the sequence of the two events.

REMINDER

Use the multiplication principle when there is a sequence of events.

The multiplication principle can be extended by induction to apply to a sequence of any finite number of events. (See Exercise 73 at the end of this section.) The multiplication principle is useful whenever we want to count the total number of possible outcomes for a task that can be broken down into a sequence of successive subtasks.

EXAMPLE 25

The last part of your telephone number contains four digits. How many such four-digit numbers are there?

We can construct four-digit numbers by performing a sequence of subtasks: choose the first digit, then the second, the third, and finally the fourth. The first digit can be any one of the 10 digits from 0 to 9, so there are 10 possible outcomes for the first subtask. Likewise, there are 10 different possibilities each for the second digit, the third, and the fourth. Using the multiplication principle, we multiply the number of outcomes for each subtask in the sequence. Therefore there are $10 \cdot 10 \cdot 10 \cdot 10 = 10,000$ different numbers. ●

If an element cannot be used again—that is, if repetitions are not allowed—the number of possible outcomes for successive events will be affected.

EXAMPLE 26

Referring to Example 25, how many four-digit numbers are there if the same digit cannot be used twice?

Again we have the sequence of subtasks of selecting the four digits, but no repetitions are allowed. There are 10 choices for the first digit, but only 9 choices for the second because we can't use what we used for the first digit, and so on. There are $10 \cdot 9 \cdot 8 \cdot 7 = 5040$ different numbers. ●

EXAMPLE 27

- How many ways are there to choose three officers from a club of 25 people?
- How many ways are there to choose three officers from a club of 25 people if someone can hold more than one office?

In (a), there are three successive subtasks with no repetitions. The first subtask, choosing the first officer, has 25 possible outcomes. The second subtask has 24 outcomes, the third 23 outcomes. The total number of outcomes is $25 \cdot 24 \cdot 23 = 13,800$. In (b), the same three subtasks are done in succession, but repetitions are allowed. The total number of outcomes is $25 \cdot 25 \cdot 25 = 15,625$. ●

EXAMPLE 28

When you order pizza at your favorite pizza place, you have the choice of thin crust, regular crust, or deep-dish; small, medium, or large; and pepperoni, sausage, barbeque, extra cheese, or vegetable. How many different pizzas can be ordered?

Again, there is a sequence of tasks: choose the crust, choose the size, and choose the topping. The total number of outcomes is $3 \cdot 3 \cdot 5 = 45$. ●

PRACTICE 22

If a man has four suits, eight shirts, and five ties, how many outfits can he put together? ■

EXAMPLE 29

For any finite set S , $|S|$ denotes the number of elements in S . If A and B are finite sets, then

$$|A \times B| = |A| \cdot |B|$$

$A \times B$ consists of all ordered pairs with first component from A and second component from B . Forming such ordered pairs can be thought of as the sequence of tasks of choosing the first component, for which there are $|A|$ outcomes, and then choosing the second component, for which there are $|B|$ outcomes. The result follows from the multiplication principle. ●

Addition Principle

Suppose we want to select a dessert from three pies and four cakes. In how many ways can this be done? There are two events, one with three outcomes (choosing a pie) and one with four outcomes (choosing a cake). However, we are not doing a sequence of two events here, since we are getting only one dessert, which must be chosen from the two disjoint sets of possibilities. The number of different outcomes is the total number of choices we have, $3 + 4 = 7$. This illustrates the *addition principle*.

● **PRINCIPLE** **ADDITION PRINCIPLE**

If A and B are disjoint events with n_1 and n_2 possible outcomes, respectively, then the total number of possible outcomes for event “ A or B ” is $n_1 + n_2$.

The addition principle can be extended by induction to the case of any finite number of disjoint events. (See Exercise 74 at the end of this section.) The addition principle is useful whenever we want to count the total number of possible outcomes for a task that can be broken down into disjoint cases.

EXAMPLE 30

A customer wants to purchase a vehicle from a dealer. The dealer has 23 autos and 14 trucks in stock. How many selections does the customer have?

REMINDER

Use the addition principle only when the events are disjoint—have no common outcomes.

The customer wants to choose a car or truck. These are disjoint events; choosing an auto has 23 outcomes and choosing a truck has 14. By the addition principle, choosing a vehicle has $23 + 14 = 37$ outcomes. Notice the requirement that the outcomes for events A and B be disjoint sets. Thus, if a customer wanted to purchase a vehicle from a dealer who had 23 autos, 14 trucks, and 17 red vehicles in stock, we could not conclude that the customer had $23 + 14 + 17$ choices! ●

EXAMPLE 31

Let A and B be disjoint finite sets. Then $|A \cup B| = |A| + |B|$.

Finding $|A \cup B|$ can be done by the disjoint cases of counting the number of elements in A , $|A|$, and the number of elements in B , $|B|$. By the addition principle, we sum these two numbers. ●

EXAMPLE 32

If A and B are finite sets, then

$$|A - B| = |A| - |A \cap B|$$

and

$$|A - B| = |A| - |B| \text{ if } B \subseteq A$$

To prove the first equality, note that

$$\begin{aligned} (A - B) \cup (A \cap B) &= (A \cap B') \cup (A \cap B) \\ &= A \cap (B' \cup B) \\ &= A \cap S \\ &= A \end{aligned}$$

so that $A = (A - B) \cup (A \cap B)$. Also, $A - B$ and $A \cap B$ are disjoint sets; therefore, by Example 31,

$$|A| = |(A - B) \cup (A \cap B)| = |A - B| + |A \cap B|$$

or

$$|A - B| = |A| - |A \cap B|$$

The second equation follows from the first, because if $B \subseteq A$ then $A \cap B = B$. ●

Using the Principles Together

Frequently the addition principle is used in conjunction with the multiplication principle.

EXAMPLE 33

Referring to Example 24, suppose we want to find how many different ways the child can *choose* the candy, rather than the number of sets of candy the child can have. Then choosing a red jellybean followed by a yellow gummy bear is not the same as choosing a yellow gummy bear followed by a red jellybean. We can consider two disjoint cases—choosing jellybeans first or choosing gummy bears first. Each of these cases (by the multiplication principle) has six outcomes, so (by the addition principle) there are $6 + 6 = 12$ possible ways to choose the candy. ●

EXAMPLE 34

How many four-digit numbers begin with a 4 or a 5?

We can consider the two disjoint cases—numbers that begin with 4 and numbers that begin with 5. Counting the numbers that begin with 4, there is 1 outcome for the subtask of choosing the first digit, then 10 possible outcomes for the subtasks of choosing each of the other three digits. Hence, by the multiplication principle there are $1 \cdot 10 \cdot 10 \cdot 10 = 1000$ ways to get a four-digit number beginning with 4. The same reasoning shows that there are 1000 ways to get a four-digit number beginning with 5. By the addition principle, there are $1000 + 1000 = 2000$ total possible outcomes. ●

PRACTICE 23

If a woman has 7 blouses, 5 skirts, and 9 dresses, how many different outfits does she have? ■

Often a counting problem can be solved in more than one way. Although the possibility of a second solution might seem confusing, it provides an excellent way to check our work; if two different ways of looking at the problem produce the same answer, it increases our confidence that we have analyzed the problem correctly.

EXAMPLE 35

Consider the problem of Example 34 again. We can avoid using the addition principle by thinking of the problem as four successive subtasks, where the first subtask, choosing the first digit, has two possible outcomes—choosing a 4 or choosing a 5. Then there are $2 \cdot 10 \cdot 10 \cdot 10 = 2000$ possible outcomes. ●

EXAMPLE 36

How many three-digit integers (numbers between 100 and 999 inclusive) are even?

One solution notes that an even number ends in 0, 2, 4, 6, or 8. Taking these as separate cases, the number of three-digit integers ending in 0 can be found by choosing the three digits in turn. There are 9 choices, 1 through 9, for the first digit; 10 choices, 0 through 9, for the second digit; and 1 choice for the third digit, 0. By the multiplication principle, there are 90 numbers ending in 0. Similarly, there are 90 numbers ending in 2, 4, 6, and 8, so by the addition principle, there are $90 + 90 + 90 + 90 + 90 = 450$ numbers.

Another solution takes advantage of the fact that there are only 5 choices for the third digit. By the multiplication principle, there are $9 \cdot 10 \cdot 5 = 450$ numbers.

For this problem, there is a third solution of the “serendipity” type we discussed in Section 2.1. There are $999 - 100 + 1 = 900$ three-digit integers in the range specified. Half are even and half are odd, so 450 of them must be even. ●

EXAMPLE 37

Suppose the last four digits of a telephone number must include at least one repeated digit. How many such numbers are there?

Although it is possible to do this problem by using the addition principle directly, it is difficult because there are so many disjoint cases to consider. For example, if the first two digits are alike but the third and fourth are different, there are $10 \cdot 1 \cdot 9 \cdot 8$ ways this can happen. If the first and third digit are alike but the second

and fourth are different, there are $10 \cdot 9 \cdot 1 \cdot 8$ ways this can happen. If the first two digits are alike and the last two are also alike but different from the first two, there are $10 \cdot 1 \cdot 9 \cdot 1$ such numbers. Obviously, there are many other possibilities.

Instead, we solve the problem by noting that numbers with repetitions and numbers with no repetitions are disjoint sets whose union equals all four-digit numbers. By Example 31 we can find the number with repetitions by subtracting the number with no repetitions (5040, according to Example 26) from the total number (10,000, according to Example 25). Therefore, there are 4960 numbers with repetitions. ●

EXAMPLE 38

A computer (or tablet, or camera, or cell phone) that wishes to connect to the Internet must have an IP (Internet Protocol) address assigned to it. This allows the device to be “found” over the Internet, much as a postal address allows a building to be “found” via regular mail. The version of IP known as IPv4 uses an address that is a 32-bit, or 4-byte, number (1 byte equals 8 bits). The first part of the address, called the netid, identifies the network that the machine is part of and the rest, called the hostid, identifies the machine itself. Note that this is a hierarchical addressing scheme. A router trying to decide where to send a data packet looks at the netid to determine the network. The hostid bytes need never be consulted until the data packet has reached the correct network. U.S. postal addresses are hierarchical in the opposite order, with the most specific information first.

How many different IPv4 addresses are there? Each of the 32 bits can be set to 0 or 1, so by the multiplication principle, there are $2 \cdot 2 \cdot 2 \cdot \dots \cdot 2 = 2^{32}$ different bit patterns. Looking at a more abstract view, assume that a particular IP address uses 16 bits for the netid and 16 bits for the hostid. Again using the multiplication principle, this would give $2^{16} \cdot 2^{16} = (\text{again}) 2^{32}$ unique IP addresses. This number is roughly 4.3 billion, which seems large enough to satisfy the world’s needs. But no — the pool of IPv4 addresses allotted to some regions of the world began to run out in 2011 and more would do so in another year or two. Hence the switch to IPv6.

An IPv6 address is 128 bits, divided into 64 bits for the network prefix that identifies a particular network and the last 64 bits for the interface ID that identifies the unique node on the network. While the gross structure of an IPv6 address therefore sounds just like an IPv4 address only bigger, there are details that make the IPv6 scheme more efficient. And exactly how big is the pool of IPv6 addresses? Using the same reasoning as before, there are 2^{128} unique addresses. This number is roughly 3.4×10^{38} , or 340 trillion trillion trillion, enough, it is said, for every star in the known universe to have the equivalent of its very own IPv4 internet.

The World IPv6 Launch occurred on June 6, 2012, but it was not like turning on a switch. Many major companies already supported IPv6 and IPv4 will continue to be supported over a few years of transition. ●

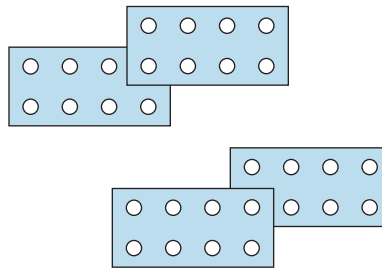
Decision Trees

Trees such as those in Figures 4.3 and 4.4 illustrate the number of outcomes of an event based on a series of possible choices. Such trees are called **decision trees**. We will see in Chapter 6 how decision trees are used in analyzing algorithms, but for now we use them to solve additional counting problems. The trees of Figures 4.3 and 4.4 led to the multiplication principle because the number of

EXERCISES 4.2

1. A frozen yogurt shop allows you to choose one flavor (vanilla, strawberry, lemon, cherry, or peach), one topping (chocolate shavings, crushed toffee, or crushed peanut brittle), and one condiment (whipped cream or shredded coconut). How many different desserts are possible?
2. In Exercise 1, how many dessert choices do you have if you are allergic to strawberries and chocolate?
3. A video game is begun by making selections from each of 3 menus. The first menu (number of players) has 4 selections, the second menu (level of play) has 8, and the third menu (speed) has 6. In how many configurations can the game be played?
4. A multiple choice exam has 20 questions each with 4 possible answers and 10 additional questions each with 5 possible answers. How many different answer sheets are possible?
5. A user's password to access a computer system consists of 3 letters followed by 2 digits. How many different passwords are possible?
6. On the computer system of Exercise 5, how many passwords are possible if uppercase and lowercase letters can be distinguished?
7. A telephone conference call is being placed from Central City to Booneville by way of Cloverdale. There are 45 trunk lines from Central City to Cloverdale and 13 from Cloverdale to Booneville. How many different ways can the call be placed?
8. A , B , C , and D are nodes on a computer network. There are 2 paths between A and C , 2 between B and D , 3 between A and B , and 4 between C and D . Along how many routes can a message from A to D be sent?
9. How many nine-digit Social Security numbers are possible?
10. An apartment building purchases a new lock system for its 175 units. A lock is opened by punching in a two-digit code. Has the apartment management made a wise purchase?
11. A palindrome is a string of characters that reads the same forward and backward. How many five-letter English language palindromes are possible?
12. How many three-digit numbers less than 600 can be made using the digits 8, 6, 4, and 2?
13. A binary logical connective can be defined by giving its truth table. How many different binary logical connectives are there?
14. Three seats on the county council are to be filled, each with someone from a different party. There are 4 candidates running from the Concerned Environmentalist party, 3 from the Limited Development party, and 2 from the Friends of the Spotted Newt party. In how many ways can the seats be filled?
15. In the original BASIC programming language, an identifier must be either a single letter or a letter followed by a single digit. How many identifiers are possible?
16. A president and vice-president must be chosen for the executive committee of an organization. There are 17 volunteers from the Eastern Division and 24 volunteers from the Western Division. If both officers must come from the same division, in how many ways can the officers be selected?
17. A dinner special allows you to select from 5 appetizers, 3 salads, 4 entrees, and 3 beverages. How many different dinners are there?
18. In Exercise 17, how many different dinners are there if you may have an appetizer or a salad but not both?
19. A new car can be ordered with a choice of 10 exterior colors; 7 interior colors; automatic or three-speed or five-speed transmission; with or without air conditioning; with or without power steering; and with or without the option package that contains the power door lock and the rear-window defroster. How many different cars can be ordered?
20. In Exercise 19, how many different cars can be ordered if the option package comes only on a car with an automatic transmission?

21. In one state, automobile license plates must have two digits (no leading zeros) followed by one letter followed by a string of two to four digits (leading zeros are allowed). How many different plates are possible?
22. A Hawaiian favorite fast food is the “loco moco,” invented at a Hilo restaurant. It consists of a bed of rice under a meat patty with egg on top, the whole thing smothered in brown gravy. The rice can be white or brown, the egg can be fried, scrambled, or poached, and the meat can be hamburger, Spam, Portuguese sausage, bacon, turkey, hot dog, salmon, or mahi. How many different loco mocos can be ordered?
23. A customer at a fast-food restaurant can order a hamburger with or without mustard, ketchup, pickle, or onion; a fish sandwich with or without lettuce, tomato, or tartar sauce; and a choice of 3 kinds of soft drinks or 2 kinds of milk shakes. How many different orders are possible if a customer can order at most 1 hamburger, 1 fish sandwich, and 1 beverage but can order less?
24. How many unique ways are there to stack two 2×4 Lego bricks of the same color? Two stacks that look the same if you merely rotate them are considered to be the same arrangement. Here is one such stack, the original on top and the same stack after rotating 180° .



25. What is the value of *Count* after the following pseudocode has been executed?

```

Count = 0
for i = 1 to 5 do
    for Letter = 'A' to 'C' do
        Count = Count + 1
    end for
end for

```

26. What is the value of *Result* after the following pseudocode has been executed?

```

Result = 0
for Index = 20 down to 10 do
    for Inner = 5 to 10 do
        Result = Result + 2
    end for
end for

```

Exercises 27–32 concern the set of three-digit integers (numbers between 100 and 999 inclusive).

27. How many are divisible by 5?
28. How many are divisible by 4?
29. How many are not divisible by 5?
30. How many are divisible by 4 or 5?
31. How many are divisible by 4 and 5?
32. How many are divisible by neither 4 nor 5?

Exercises 33–42 concern the set of binary strings of length 8 (each character is either the digit 0 or the digit 1).

33. How many such strings are there?
34. How many begin and end with 0?
35. How many begin or end with 0?
36. How many have 1 as the second digit?
37. How many begin with 111?
38. How many contain exactly one 0?
39. How many begin with 10 or have a 0 as the third digit?
40. How many are palindromes? (See Exercise 11.)
41. How many contain exactly seven 1s?
42. How many contain two or more 0s?

In Exercises 43–48, 2 dice are rolled, 1 black and 1 white.

43. How many different rolls are possible? (Note that a 4-black, 1-white result and a 1-black, 4-white result are two different outcomes.)
44. How many rolls result in “snake eyes” (both dice showing 1)?
45. How many rolls result in doubles (both dice showing the same value)?
46. How many rolls result in a total of 7 or 11?
47. How many rolls occur in which neither die shows the value 4?
48. How many rolls occur in which the value on the white die is greater than the value on the black die?

In Exercises 49–54, a customer is ordering a new desktop computer system. The choices are 21-inch, 23-inch, or 24-inch monitor (optional); 1 TB or 2 TB hard drive; 6 GB or 8 GB of RAM; 16X DVD or Blu-ray disk optical drive; Intel, AMD, or NVIDIA video card; inkjet, laser, or laser color printer; 1-, 2-, or 3-year warranty.

49. How many different machine configurations are possible?
50. How many different machines can be ordered with a 2 TB hard drive?
51. How many different machines can be ordered with a 21-inch monitor and an inkjet printer?
52. How many different machines can be ordered if the customer decides not to get a new monitor?
53. How many different machines can be ordered with a 21-inch monitor, a 1 TB hard drive, and an inkjet printer?
54. How many different machines can be ordered if the customer does not want a 3-year warranty?

Exercises 55–58 refer to the Konami code, a sequence of 10 keystrokes using 6 different characters: $\uparrow \downarrow \downarrow \leftarrow \rightarrow \leftarrow \rightarrow$ B A. This code was inadvertently left by the developer in the first release of a video game; when entered on the video game console as the title screen is open, it surreptitiously adds assets to the player’s on-screen avatar. This code or versions of it have been retained in many video games.

55. How many 10-character codes can be created using these 6 characters?
56. How many 10-character codes can be created if the BA sequence must be the last 2 characters?
57. How many 10-character codes can be created if only 2 down-arrow characters can be used and they must be paired together wherever they appear?
58. How many 10-character codes can be created if only the arrow characters are used?

In Exercises 59–68, a hand consists of 1 card drawn from a standard 52-card deck with flowers on the back and 1 card drawn from a standard 52-card deck with birds on the back. A standard deck has 13 cards from each of 4 suits (clubs, diamonds, hearts, spades). The 13 cards have face value 2 through 10, jack, queen, king, or ace. Each face value is a “kind” of card. The jack, queen, and king are “face cards.”

59. How many different hands are possible? (Note that a flower-ace-of-spades, bird-queen-of-hearts and a flower-queen-of-hearts, bird-ace-of-spades are two different outcomes.)
60. How many hands consist of a pair of aces?
61. How many hands contain all face cards?
62. How many hands contain exactly 1 king?
63. How many hands consist of two of a kind (2 aces, 2 jacks, and so on)?
64. How many hands have a face value of 5 (aces count as 1, face cards count as 10)?
65. How many hands have a face value of less than 5 (aces count as 1, face cards count as 10)?
66. How many hands do not contain any face cards?
67. How many hands contain at least 1 face card?
68. How many hands contain at least 1 king?
69. Draw a decision tree to find the number of binary strings of length 4 that do not have consecutive 0s. (Compare your answer with the one for Exercise 41 of Section 3.2.)
70. Draw a decision tree (use teams A and B) to find the number of ways the NBA playoffs can happen, where the winner is the first team to win 4 out of 7.
71. Voting on a certain issue is conducted by having everyone put a red, blue, or green slip of paper into a hat. Then the slips are pulled out one at a time. The first color to receive two votes wins. Draw a decision tree to find the number of ways in which the balloting can occur.
72. In Example 39, prove the following facts, where $C(n)$ = the total number of nodes in the decision tree at level n , $H(n)$ = the total number of nodes at level n resulting from an H toss, $T(n)$ = the total number of nodes at level n resulting from a T toss.¹
 - a. $C(n) = H(n) + T(n)$
 - b. $H(n) = T(n - 1)$
 - c. $T(n) = H(n - 1) + T(n - 1)$
 - d. $H(n) = H(n - 2) + T(n - 2)$
 - e. $C(n) = C(n - 2) + C(n - 1)$ for $n \geq 3$
 - f. $C(n) = F(n + 1)$ where $F(n)$ is the n th Fibonacci number
73. Use mathematical induction to extend the multiplication principle to a sequence of m events for any integer m , $m \geq 2$.
74. Use mathematical induction to extend the addition principle to m disjoint events for any integer m , $m \geq 2$.
75. Consider the product of n factors, $x_1 \cdot x_2 \cdot \cdots \cdot x_n$. Such an expression can be fully parenthesized to indicate the order of multiplication in a number of ways. For example, if $n = 4$, there are five ways to parenthesize:

$$\begin{aligned}
 &x_1 \cdot (x_2 \cdot (x_3 \cdot x_4)) \\
 &x_1 \cdot ((x_2 \cdot x_3) \cdot x_4) \\
 &(x_1 \cdot x_2) \cdot (x_3 \cdot x_4) \\
 &(x_1 \cdot (x_2 \cdot x_3)) \cdot x_4 \\
 &((x_1 \cdot x_2) \cdot x_3) \cdot x_4
 \end{aligned}$$

¹This problem was suggested by Mr. Tracy Castile, a former University of Hawaii at Hilo student.

- a. Write a recurrence relation for $P(n)$, the number of ways to parenthesize a product of n factors, $n \geq 1$. Assume that $P(1) = 1$. (Hint: Note that for $n > 2$, the last multiplication to be performed can occur in any of $n - 1$ positions.)
- b. Prove that

$$P(n) = C(n - 1)$$

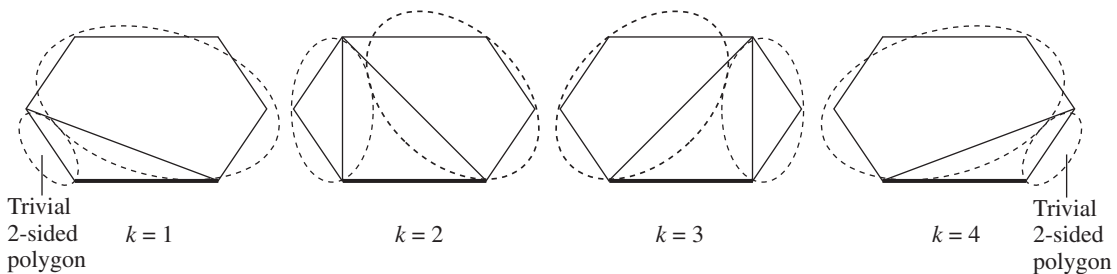
where $C(0), C(1), \dots$ is the sequence of Catalan numbers (see Exercise 38 of Section 3.1).

76. A simple closed convex polygon consists of n points in the plane joined in pairs by n line segments; each point is the endpoint of exactly 2 line segments, and any line connecting 2 nonadjacent points lies wholly within the polygon.
- a. Show that an $(n + 2)$ -sided simple closed convex polygon can be triangulated (divided into triangular regions) using $n - 1$ lines. (The figures show two different triangulations of a 6-sided polygon, where $n = 4$).



(Hint: Use a pair of straight lines to shave off 2 corners; consider the cases where n is even and the cases where n is odd.)

- b. Write a recurrence relation for $T(n)$, the number of different triangulations of an $(n + 2)$ -sided polygon. Assume that $T(0) = 1$. (Hint: Fix one edge of the polygon as the base of a triangle whose tip rotates around the polygon, as shown. Use the sides of the triangle to divide the polygon into 2 polygonal sections with $(k + 1)$ sides and $(n - k + 2)$ sides.)



- c. Prove that $T(n) = C(n)$, where $C(0), C(1), \dots$ is the sequence of Catalan numbers (see Exercise 38 of Section 3.1).

SECTION 4.3 PRINCIPLE OF INCLUSION AND EXCLUSION; PIGEONHOLE PRINCIPLE

In this section we discuss two more counting principles that can be used to solve combinatorics problems.

Relations, Functions, and Matrices

CHAPTER OBJECTIVES

After studying this chapter, you will be able to:

- Identify ordered pairs related by a binary relation.
- Test a binary relation for the reflexive, symmetric, transitive, and antisymmetric properties.
- Find the reflexive, symmetric, and transitive closures of a binary relation.
- Recognize partial orderings and construct Hasse diagrams for them.
- Recognize an equivalence relation on a set and describe how it partitions the set into equivalence classes.
- Draw a PERT chart from a task table.
- Find the minimum time-to-completion and a critical path in a PERT chart.
- Extend a partial ordering on a finite set to a total ordering by doing a topological sort.
- Understand the entity-relationship model and the relational model for an enterprise.
- Perform restrict, project, and join operations in a relational database.
- Create relational database queries in the languages of relational algebra, SQL, and relational calculus.
- Determine whether a binary relation is a function.
- Test a function for the onto and one-to-one properties.
- Create composite functions.
- Decide whether a function has an inverse function and what the inverse function is.
- Manipulate cycle notation for permutation functions.
- Compute the number of functions, onto functions, and one-to-one functions from one finite set to another.
- Understand order of magnitude as a relative measure of function rate of growth.
- Build a hash table using a modulo hash function.
- Encode and decode messages using RSA public key encryption.
- Use the mod function to compute check digits for various identification codes.
- Perform matrix arithmetic on matrices of appropriate dimensions.
- Solve systems of linear equations using Gaussian elimination.
- Perform Boolean arithmetic operations on Boolean matrices of appropriate dimensions.

Your company has developed a program for use on a small parallel processing machine. According to the technical documentation, the program executes processes P1, P2, and P3 in parallel; these processes all need results from process P4, so they must wait for Process P4 to complete execution before they begin. Processes P7 and P10 execute in parallel but must wait until processes P1, P2, and P3 have finished. Process P4 requires results from P5 and P6 before it can begin execution. P5 and P6 execute in parallel. Processes P8 and P11 execute in parallel, but P8 must wait for Process P7 to complete and P11 must wait for process P10 to complete. Process P9 must wait for results from P8 and P11. You have been assigned to convert the software for use on a single processor machine.

Question: In what order should the processes be executed?

Here various pairs of processes are related to one another by a “prerequisite” relation. This is a special case of a binary relation, a relationship between pairs of elements within a set. We will study the various properties of binary relations in Section 5.1. One type of binary relation is called a partial ordering; elements related by a partial ordering can be represented graphically. Another type of binary relation is an equivalence relation; elements related by an equivalence relation can be grouped into classes.

A topological sort extends a partial ordering to a total ordering. For a partial ordering of prerequisite tasks, a corresponding total ordering identifies the sequential order in which the tasks would have to be done, which is the solution to the parallel processing conversion problem. Topological sorting is presented in Section 5.2.

A generalization of a binary relation forms the basis for relational databases, considered in Section 5.3. Using operations of restrict, project, and join on the relations in a database, we can make various queries of the database.

A function is a special kind of binary relation. Functions, as well as relations, describe a number of real-world situations. Functions can also have special properties, as discussed in Section 5.4. Order of magnitude, presented in Section 5.5, provides a way to compare the growth rate of two functions and is useful in the analysis of algorithms. A simple function called the modulo function has a surprising number of applications, ranging from encryption algorithms for computer security to the basis for artistic design patterns. Some of these applications are mentioned in Section 5.6.

In Section 5.7, we consider matrices and develop an arithmetic for manipulating them. Matrices provide a mechanism for solving systems of linear equations. We will later use matrices to represent relations and graphs.

SECTION 5.1 RELATIONS

Binary Relations

If we learn that two people, Henrietta and Horace, are related, we understand that there is some family connection between them—that (Henrietta, Horace) stands out from other ordered pairs of people because there is a relationship (cousins, sister and brother, or whatever) that Henrietta and Horace satisfy. The mathematical

analogue is to distinguish certain ordered pairs of objects from other ordered pairs because the components of the distinguished pairs satisfy some relationship that the components of the other pairs do not.

EXAMPLE 1

Remember (Section 4.1) that the Cartesian product of a set S with itself, $S \times S$ or S^2 , is the set of all ordered pairs of elements of S . Let $S = \{1, 2, 3\}$; then

$$S \times S = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$$

If we were interested in the relationship of equality, then $(1, 1)$, $(2, 2)$, $(3, 3)$ would be the distinguished elements of $S \times S$, that is, the only ordered pairs whose components are equal. If we were interested in the relationship of one number being less than another, we would choose $(1, 2)$, $(1, 3)$, and $(2, 3)$ as the distinguished ordered pairs of $S \times S$. •

In Example 1, we could pick out the distinguished ordered pairs (x, y) by saying that $x = y$ or that $x < y$. Similarly, the notation $x \rho y$ indicates that the ordered pair (x, y) satisfies a relation ρ . The relation ρ may be defined in words or by an equation or simply by listing the ordered pairs that satisfy ρ .

EXAMPLE 2

Let $S = \{1, 2, 4\}$. On the set $S \times S = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (2, 4), (4, 1), (4, 2), (4, 4)\}$, a relation ρ can be defined by $x \rho y$ if and only if $x = y/2$, abbreviated $x \rho y \leftrightarrow x = y/2$. Thus $(1, 2)$ and $(2, 4)$ satisfy ρ . Alternatively, the same ρ could be defined by saying that $\{(1, 2), (2, 4)\}$ is the set of ordered pairs satisfying ρ . •

As in Example 2, one way to define the binary relation ρ is to specify a subset of $S \times S$. Formally, this is the definition of a binary relation on a set.

• **DEFINITION BINARY RELATION ON A SET S**

A **binary relation on a set S** is a subset of $S \times S$ (a set of ordered pairs of elements of S).

Now that we know that a binary relation ρ is a subset, we see that

$$x \rho y \leftrightarrow (x, y) \in \rho$$

Generally, a binary relation is defined by describing the relation rather than by listing the ordered pairs. The description gives a characterizing property of elements of the relation; that is, it is a binary predicate satisfied by certain ordered pairs. A binary relation implies a yes/no result—an ordered pair either does or does not satisfy the binary predicate and either does or does not belong to the relation.

EXAMPLE 3

Let $S = \{1, 2\}$. Then $S \times S = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$. Let ρ on S be given by the description $x \rho y \leftrightarrow x + y$ is odd. Then $(1, 2) \in \rho$ and $(2, 1) \in \rho$. The ordered pair $(1, 1) \notin \rho$ because $1 + 1$ is not odd. Similarly $(2, 2) \notin \rho$. ●

EXAMPLE 4

Let $S = \{1, 2\}$. Then $S \times S = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$. If ρ is defined on S by $\rho = \{(1, 1), (2, 1)\}$, then $1 \rho 1$ and $2 \rho 1$ hold, but not, for instance, $1 \rho 2$. Here ρ seems to have no obvious verbal description. ●

In this section we will be concerned almost exclusively with binary relations on a single set, but more generally, relations can be defined on multiple sets.

● **DEFINITION** **RELATIONS ON MULTIPLE SETS**

Given two sets S and T , a **binary relation from S to T** (also called a **binary relation on $S \times T$**) is a subset of $S \times T$. Given n sets $S_1, S_2, \dots, S_n$, $n > 2$, an **n -ary relation on $S_1 \times S_2 \times \dots \times S_n$** is a subset of $S_1 \times S_2 \times \dots \times S_n$.

EXAMPLE 5

Let $S = \{1, 2, 3\}$ and $T = \{2, 4, 7\}$. Then the set

$$\{(1, 2), (2, 4), (2, 7)\}$$

consists of elements from $S \times T$. It is a binary relation from S to T . ●

PRACTICE 1

For each of the following binary relations ρ on $\mathbb{N}$, decide which of the given ordered pairs belong to ρ .

- $x \rho y \leftrightarrow x = y + 1$; $(2, 2), (2, 3), (3, 3), (3, 2)$
- $x \rho y \leftrightarrow x$ divides y ; $(2, 4), (2, 5), (2, 6)$
- $x \rho y \leftrightarrow x$ is odd; $(2, 3), (3, 4), (4, 5), (5, 6)$
- $x \rho y \leftrightarrow x > y^2$; $(1, 2), (2, 1), (5, 2), (6, 4), (4, 3)$

If ρ is a binary relation on S , then ρ will consist of a set of ordered pairs of the form (s_1, s_2) . A given first component s_1 or second component s_2 can be paired in various ways in the relation. The relation is **one-to-one** if each first component and each second component appears only once in the relation. The relation is **one-to-many** if some first component s_1 appears more than once; that is, one s_1 is paired with more than one second component. It is **many-to-one** if some second component s_2 is paired with more than one first component. Finally, it is **many-to-many** if at least one s_1 is paired with more than one second component and at least one s_2 is paired with more than one first component. Figure 5.1 illustrates these four possibilities. Note that not all values in S need be components in ordered pairs of ρ .

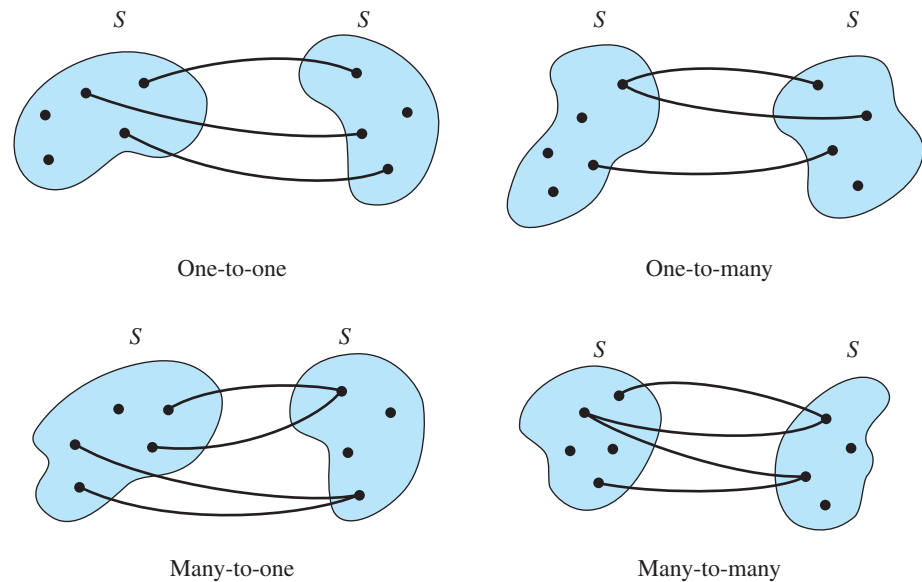


Figure 5.1

These ideas extend to relations from a set S to a set T . The relation of Example 5 is one-to-many because the first component 2 appears more than once; 2 from set S is associated with both 4 and 7 from set T .

PRACTICE 2 Identify each of these relations on S , where $S = \{2, 5, 7, 9\}$, as one-to-one, one-to-many, many-to-one, or many-to-many.

- $\{(5, 2), (7, 5), (9, 2)\}$
- $\{(2, 5), (5, 7), (7, 2)\}$
- $\{(7, 9), (2, 5), (9, 9), (2, 7)\}$

Suppose B is the set of all binary relations on a given set S . If ρ and σ belong to B , then they are subsets of $S \times S$. As such, we can perform set operations of union, intersection, and complementation that result in new subsets of $S \times S$, that is, new binary relations, which we will denote by $\rho \cup \sigma$, $\rho \cap \sigma$, and ρ' , respectively. Thus

$$\begin{aligned} x(\rho \cup \sigma)y &\leftrightarrow x\rho y \text{ or } x\sigma y \\ x(\rho \cap \sigma)y &\leftrightarrow x\rho y \text{ and } x\sigma y \\ x\rho'y &\leftrightarrow \text{not } x\rho y \end{aligned}$$

PRACTICE 3 Let ρ and σ be two binary relations on $\mathbb{N}$ defined by $x\rho y \leftrightarrow x = y$ and $x\sigma y \leftrightarrow x < y$. Give verbal descriptions for parts (a), (b), and (c); give a set description for part (d).

- What is the relation $\rho \cup \sigma$?
- What is the relation ρ' ?
- What is the relation σ' ?
- What is the relation $\rho \cap \sigma$?

The following facts about the operations $\cup$, $\cap$ and $'$ on relations are immediate consequences of the basic set identities found in Section 4.1. The set S^2 (which is, after all, a subset of S^2) is being viewed here as a binary relation on S .

$$\begin{array}{ll} \text{1a. } \rho \cup \sigma = \sigma \cup \rho & \text{1b. } \rho \cap \sigma = \sigma \cap \rho \\ \text{2a. } (\rho \cup \sigma) \cup \gamma = \rho \cup (\sigma \cup \gamma) & \text{2b. } (\rho \cap \sigma) \cap \gamma = \rho \cap (\sigma \cap \gamma) \\ \text{3a. } \rho \cup (\sigma \cap \gamma) = (\rho \cup \sigma) \cap (\rho \cup \gamma) & \text{3b. } \rho \cap (\sigma \cup \gamma) = (\rho \cap \sigma) \cup (\rho \cap \gamma) \\ \text{4a. } \rho \cup \emptyset = \rho & \text{4b. } \rho \cap S^2 = \rho \\ \text{5a. } \rho \cup \rho' = S^2 & \text{5b. } \rho \cap \rho' = \emptyset \end{array}$$

REMINDER

Reflexive—Every x is related to itself.

Symmetric—If x is related to y , then y is related to x .

Transitive—If x is related to y and y is related to z , then x is related to z .

Properties of Relations

A binary relation on a set S may have certain properties. For example, the relation ρ of equality on S , $(x, y) \in \rho \leftrightarrow x = y$, has three properties: (1) for any $x \in S$, $x = x$, or $(x, x) \in \rho$; (2) for any $x, y \in S$, if $x = y$ then $y = x$, or $(x, y) \in \rho \rightarrow (y, x) \in \rho$; and (3) for any $x, y, z \in S$, if $x = y$ and $y = z$, then $x = z$, or $[(x, y) \in \rho \text{ and } (y, z) \in \rho] \rightarrow (x, z) \in \rho$. These three properties make the equality relation reflexive, symmetric, and transitive.

DEFINITION REFLEXIVE, SYMMETRIC, AND TRANSITIVE RELATIONS

Let ρ be a binary relation on a set S . Then

$$\begin{array}{ll} \rho \text{ is } \mathbf{reflexive} \text{ means} & (\forall x)(x \in S \rightarrow (x, x) \in \rho) \\ \rho \text{ is } \mathbf{symmetric} \text{ means} & (\forall x)(\forall y)(x \in S \wedge y \in S \wedge (x, y) \in \rho \rightarrow (y, x) \in \rho) \\ \rho \text{ is } \mathbf{transitive} \text{ means} & (\forall x)(\forall y)(\forall z)(x \in S \wedge y \in S \wedge z \in S \wedge \\ & (x, y) \in \rho \wedge (y, z) \in \rho \rightarrow (x, z) \in \rho) \end{array}$$

EXAMPLE 6

Consider the relation $\leq$ on the set $\mathbb{N}$. This relation is reflexive because for any nonnegative integer x , $x \leq x$. It is also a transitive relation because for any nonnegative integers x , y , and z , if $x \leq y$ and $y \leq z$, then $x \leq z$. However, $\leq$ is not symmetric; $3 \leq 4$ does not imply $4 \leq 3$. In fact, for any $x, y \in \mathbb{N}$, if both $x \leq y$ and $y \leq x$, then $x = y$. This characteristic is described by saying that $\leq$ is antisymmetric. ●

DEFINITION ANTISYMMETRIC RELATION

Let ρ be a binary relation on a set S . Then ρ is **antisymmetric** means

$$(\forall x)(\forall y)(x \in S \wedge y \in S \wedge (x, y) \in \rho \wedge (y, x) \in \rho \rightarrow x = y)$$

EXAMPLE 7

Let $S = \wp(\mathbb{N})$. Define a binary relation ρ on S by $A \rho B \leftrightarrow A \subseteq B$. Then ρ is reflexive because every set is a subset of itself. Also, ρ is transitive because if A is a subset of B and B is a subset of C , then A is a subset of C . Finally, ρ is antisymmetric because if A is a subset of B and B is a subset of A , then A and B are equal sets. ●

REMINDER

Antisymmetric—If x is related to y and y is related to x , then $x = y$.

All four relational properties involve the implication connective. The universal quantifiers mean that the implications must be true for arbitrary choices of variables. Recall that to prove an implication true, we assume that the antecedent is true and prove that the consequent must also be true. For the reflexive property, the antecedent just chooses an arbitrary element in S ; the consequent says that this element must be related to itself. For a relation ρ on a set to be reflexive, then, every element in the set must be related to itself, which specifies certain ordered pairs that must belong to ρ .

However, in the symmetric, transitive, and antisymmetric properties, the antecedent does not say only that the elements are in S . To prove that a relation is symmetric, for example, we must show that if x and y are arbitrary elements in S and if in addition x is related to y , then it must be the case that y is related to x . This says that if certain ordered pairs are found in ρ , then certain other ordered pairs must also be in ρ in order for ρ to be symmetric. In other words, knowledge of the set S is critical to determining whether reflexivity holds, while to determine the other properties, it is sufficient just to look at the ordered pairs in ρ .

At any rate, the question of whether a given relation on a set S has a certain property requires a yes or no answer. The property either holds or it doesn't.

PRACTICE 4 Let $S = \{1, 2, 3\}$.

- If a relation ρ on S is reflexive, what ordered pairs must belong to ρ ?
- If a relation ρ on S is symmetric, what ordered pairs must belong to ρ ? (This is a trick question; see the answer at the back of the book.)
- If a relation ρ on S is symmetric and if $(a, b) \in \rho$, then what other ordered pair must belong to ρ ?
- If a relation ρ on S is antisymmetric and if (a, b) and (b, a) belong to ρ , what must be true?
- Is the relation $\rho = \{(1, 2)\}$ on S transitive? (*Hint*: Remember the truth table for implication.) ■

The properties of symmetry and antisymmetry for binary relations are not precisely opposites. Antisymmetric does not mean “not symmetric.” A relation is not symmetric if some (x, y) belongs to the relation but (y, x) does not. More formally, not symmetric means

$$\begin{aligned} & ((\forall x)(\forall y)[x \in S \wedge y \in S \wedge (x, y) \in \rho \rightarrow (y, x) \in \rho])' \\ & \leftrightarrow (\exists x)(\exists y)[x \in S \wedge y \in S \wedge (x, y) \in \rho \rightarrow (y, x) \notin \rho] \\ & \leftrightarrow (\exists x)(\exists y)[(x \in S \wedge y \in S \wedge (x, y) \in \rho)' \vee (y, x) \in \rho] \\ & \leftrightarrow (\exists x)(\exists y)[(x \in S \wedge y \in S \wedge (x, y) \in \rho) \wedge (y, x) \notin \rho] \end{aligned}$$

Relations can therefore be symmetric and not antisymmetric, antisymmetric and not symmetric, both, or neither.

The equality relation on a set S is both symmetric and antisymmetric. However, the equality relation on S (or a subset of this relation) is the only relation having both these properties. To illustrate, suppose ρ is a symmetric and antisymmetric relation on S , and let $(x, y) \in \rho$. By symmetry, it follows that $(y, x) \in \rho$. But by antisymmetry, $x = y$. Thus, only equal elements can be related. The relation $\rho = \{(1, 2), (2, 1), (1, 3)\}$ on the set $S = \{1, 2, 3\}$ is neither symmetric— $(1, 3)$ belongs but $(3, 1)$ does not—nor antisymmetric— $(1, 2)$ and $(2, 1)$ belong, but $1 \neq 2$.

PRACTICE 5

Test each binary relation on the given set S for reflexivity, symmetry, antisymmetry, and transitivity.

- $S = \mathbb{N}$; $x \rho y \leftrightarrow x + y$ is even
- $S = \mathbb{Z}^+$ (positive integers); $x \rho y \leftrightarrow x$ divides y
- $S =$ set of all lines in the plane; $x \rho y \leftrightarrow x$ is parallel to y or x coincides with y
- $S = \mathbb{N}$; $x \rho y \leftrightarrow x = y^2$
- $S = \{0, 1\}$; $x \rho y \leftrightarrow x = y^2$
- $S = \{x \mid x \text{ is a person living in Peoria}\}$; $x \rho y \leftrightarrow x$ is older than y
- $S = \{x \mid x \text{ is a student in your class}\}$; $x \rho y \leftrightarrow x$ sits in the same row as y
- $S = \{1, 2, 3\}$; $\rho = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$

EXAMPLE 8

The discussion on recursion in Prolog (Section 1.5) noted that a recursive rule should be used when the predicate being described is one that is inherited from one object to the next. The predicate *in-food-chain* used there has this property because

$$\text{in-food-chain}(x, y) \wedge \text{in-food-chain}(y, z) \rightarrow \text{in-food-chain}(x, z)$$

Now we see that this is simply the transitive property.

Closures of Relations

If a relation ρ on a set S fails to have a certain property, we may be able to extend ρ to a relation ρ^* on S that does have that property. By “extend,” we mean that the new relation ρ^* will contain all the ordered pairs in ρ plus the additional ordered pairs needed for the desired property to hold. Thus $\rho \subseteq \rho^*$. If ρ^* is the smallest such set, then ρ^* is called the **closure** of ρ with respect to that property.

● **DEFINITION CLOSURE OF A RELATION**

A binary relation ρ^* on a set S is the **closure of a relation** ρ on S with respect to property P if

- ρ^* has property P .
- $\rho \subseteq \rho^*$.
- ρ^* is a subset of any other relation on S that includes ρ and has property P .

We can look for the **reflexive closure**, the **symmetric closure**, and the **transitive closure** of a relation on a set. Of course, if the relation already has a property, it is its own closure with respect to that property.

EXAMPLE 9

Let $S = \{1, 2, 3\}$ and $\rho = \{(1, 1), (1, 2), (1, 3), (3, 1), (2, 3)\}$. Then ρ is not reflexive, not symmetric, and not transitive. The closure of ρ with respect to reflexivity is

$$\rho^* = \{(1, 1), (1, 2), (1, 3), (3, 1), (2, 3), (2, 2), (3, 3)\}$$

This relation is reflexive and contains ρ . Furthermore, any reflexive relation on S would have to contain the new ordered pairs we've added— $(2, 2)$ and $(3, 3)$ —so no smaller reflexive relation can exist; that is, any reflexive relation containing ρ must have the relation above as a subset.

The closure of ρ with respect to symmetry is

$$\rho^* = \{(1, 1), (1, 2), (1, 3), (3, 1), (2, 3), (2, 1), (3, 2)\}$$

Here it is also clear that we have added just those new pairs required— $(2, 1)$ and $(3, 2)$ —for the relation to be symmetric.

For both reflexive closure and symmetric closure, we only had to inspect the ordered pairs already in ρ to find out what ordered pairs we needed to add (assuming we knew what the set S was). The reflexive or symmetric closure of the relation could be found in one step. Transitive closure may require a series of steps. Inspecting the ordered pairs in our example ρ , we see that we need to add $(3, 2)$ (because of $(3, 1)$ and $(1, 2)$), $(3, 3)$ (because of $(3, 1)$ and $(1, 3)$), and $(2, 1)$ (because of $(2, 3)$ and $(3, 1)$). This gives the relation

$$\{(1, 1), (1, 2), (1, 3), (3, 1), (2, 3), (3, 2), (3, 3), (2, 1)\}$$

However, this relation is still not transitive. Because of the new pair $(2, 1)$ and the old pair $(1, 2)$, we need to add $(2, 2)$. This gives the relation

$$\{(1, 1), (1, 2), (1, 3), (3, 1), (2, 3), (3, 2), (3, 3), (2, 1), (2, 2)\}$$

which is transitive and also the smallest transitive relation containing ρ . It is the transitive closure of ρ . ●

As in Example 9, one way to find the transitive closure of a relation is to inspect the ordered pairs in the original relation, add new pairs if necessary, inspect the resulting relation, add new pairs if necessary, and so on, until a transitive relation is achieved. This is a rather ad hoc procedure, and we will give a better algorithm in Chapter 7. There we will also see that the transitive closure of a binary relation is related to “reachability in a directed graph,” which has many applications.

PRACTICE 6 Does it make sense to look for the antisymmetric closure of a relation on a set? Why or why not? ■

PRACTICE 7 Find the reflexive, symmetric, and transitive closure of the relation

$$\{(a, a), (b, b), (c, c), (a, c), (a, d), (b, d), (c, a), (d, a)\}$$

on the set $S = \{a, b, c, d\}$. ■

For the rest of this section we will concentrate on two types of binary relations that are characterized by which properties (reflexivity, symmetry, antisymmetry, and transitivity) they satisfy.

Partial Orderings

DEFINITION PARTIAL ORDERING

A binary relation on a set S that is reflexive, antisymmetric, and transitive is called a **partial ordering** on S .

From previous examples and Practice 5, we have the following instances of partial orderings:

- On $\mathbb{N}$, $x \rho y \leftrightarrow x \leq y$.
- On $\wp(\mathbb{N})$, $A \rho B \leftrightarrow A \subseteq B$.
- On $\mathbb{Z}^+$, $x \rho y \leftrightarrow x$ divides y .
- On $\{0, 1\}$, $x \rho y \leftrightarrow x = y^2$.

If ρ is a partial ordering on S , then the ordered pair (S, ρ) is called a **partially ordered set** (also known as a **poset**). We will denote an arbitrary, partially ordered set by $(S, \leq)$; in any particular case, $\leq$ has some definite meaning such as “less than or equal to,” “is a subset of,” “divides,” and so on. (The symbol for a generic partial ordering, $\leq$, is designed to resemble the inequality symbol $\leq$, which, as we’ve just noted, is a partial ordering on the set $\mathbb{N}$ or on any other set in which a less-than-or-equal-to relation makes sense.)

Let $(S, \leq)$ be a partially ordered set, and let $A \subseteq S$. Then $\leq$ is a set of ordered pairs of elements of S , some of which may be ordered pairs of elements of A . If we select from $\leq$ the ordered pairs of elements of A , this new set is called the **restriction** of $\leq$ to A and is a partial ordering on A . (Do you see why the three required properties still hold?) For instance, once we know that the relation “ x divides y ” is a partial ordering on $\mathbb{Z}^+$, we automatically know that “ x divides y ” is a partial ordering on $\{1, 2, 3, 6, 12, 18\}$.

We want to introduce some terminology about partially ordered sets. Let $(S, \leq)$ be a partially ordered set. If $x \leq y$, then either $x = y$ or $x \neq y$. If $x \leq y$ but $x \neq y$, we write $x < y$ and say that x is a **predecessor** of y or y is a **successor** of x . A given y may have many predecessors, but if $x < y$ and there is no z with $x < z < y$, then x is an **immediate predecessor** of y .

PRACTICE 8 Consider the relation “ x divides y ” on $\{1, 2, 3, 6, 12, 18\}$.

- a. Write the ordered pairs (x, y) of this relation.
- b. Write all the predecessors of 6.
- c. Write all the immediate predecessors of 6.

If S is finite, we can visually depict a partially ordered set $(S, \leq)$ by using a **Hasse diagram**. Each of the elements of S is represented by a dot, called a

node, or **vertex**, of the diagram. If x is an immediate predecessor of y , then the node for y is placed above the node for x and the two nodes are connected by a straight-line segment.

EXAMPLE 10

Consider $\wp(\{1, 2\})$ under the relation of set inclusion. This is a partially ordered set, a restriction of the partially ordered set $(\wp(\mathbb{N}), \subseteq)$. The elements of $\wp(\{1, 2\})$ are $\emptyset$, $\{1\}$, $\{2\}$, and $\{1, 2\}$. The binary relation $\subseteq$ consists of the following ordered pairs:

$$(\emptyset, \emptyset), (\{1\}, \{1\}), (\{2\}, \{2\}), (\{1, 2\}, \{1, 2\}), (\emptyset, \{1\}), (\emptyset, \{2\}), (\emptyset, \{1, 2\}), (\{1\}, \{1, 2\}), (\{2\}, \{1, 2\})$$

The Hasse diagram of this partially ordered set appears in Figure 5.2. Note that although $\emptyset$ is not an immediate predecessor of $\{1, 2\}$, it is a predecessor of $\{1, 2\}$ (shown in the diagram by the chain of upward line segments connecting $\emptyset$ with $\{1, 2\}$).

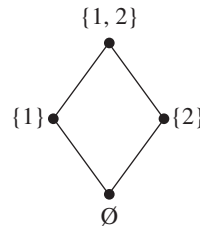


Figure 5.2

REMINDER

Two nodes in a Hasse diagram should never be joined by a horizontal line.

PRACTICE 9

Draw the Hasse diagram for the relation “ x divides y ” on $\{1, 2, 3, 6, 12, 18\}$.

The Hasse diagram of a partially ordered set conveys all the information about the partial ordering. We can reconstruct the set of ordered pairs making up the partial ordering just by looking at the diagram. The lines in the diagram tell us immediate (predecessor, successor) pairs. We can fill in the rest by using the reflexive and transitive properties. Thus, given the Hasse diagram in Figure 5.3 of a partial ordering $\leq$ on a set $\{a, b, c, d, e, f\}$, we can conclude that $\leq$ is the set

$$\{(a, a), (b, b), (c, c), (d, d), (e, e), (f, f), (a, b), (a, c), (a, d), (a, e), (d, e)\}$$

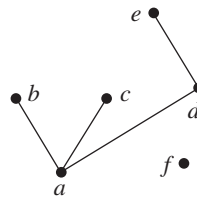


Figure 5.3

Two elements of S may be unrelated in a partial ordering of S . In Example 10, $\{1\}$ and $\{2\}$ are unrelated; so are 2 and 3, and 12 and 18 in Practice 9. In Figure 5.3, f is not related to any other element. A partial ordering in which every element of the set is related to every other element is called a **total ordering**, or **chain**. The Hasse diagram for a total ordering on a four-element set looks like Figure 5.4. The relation $\leq$ on $\mathbb{N}$ is a total ordering, although we can't draw a Hasse diagram because $\mathbb{N}$ is an infinite set.



Figure 5.4

Again, let $(S, \leq)$ be a partially ordered set. If there is a $y \in S$ with $y \leq x$ for all $x \in S$, then y is a **least element** of the partially ordered set. A least element, if it exists, is unique. To show its uniqueness, assume that y and z are both least elements. Then $y \leq z$ because y is least and $z \leq y$ because z is least; by antisymmetry, $y = z$. An element $y \in S$ is **minimal** if there is no $x \in S$ with $x < y$. In the Hasse diagram, a least element is below all others, while a minimal element has no elements below it. Similar definitions apply for greatest element and maximal elements.

PRACTICE 10 Define **greatest element** and **maximal element** in a partially ordered set $(S, \leq)$.

EXAMPLE 11

In the partially ordered set of Practice 9, 1 is both least and minimal; 12 and 18 are both maximal, but there is no greatest element.

A least element is always minimal and a greatest element is always maximal, but the converses are not true (see Example 11). In a totally ordered set, however, a minimal element is the least element and a maximal element is the greatest element.

PRACTICE 11 Draw the Hasse diagram for a partially ordered set with four elements in which there are two minimal elements but no least element, two maximal elements but no greatest element, and each element is related to exactly two other elements.

Partial orderings satisfy the properties of reflexivity, antisymmetry, and transitivity. Another type of binary relation, which we study next, satisfies a different set of properties.

Equivalence Relations

DEFINITION EQUIVALENCE RELATION

A binary relation on a set S that is reflexive, symmetric, and transitive is called an **equivalence relation** on S .

REMINDER

A partial ordering is anti-symmetric; an equivalence relation is symmetric.

We have already come across the following examples of equivalence relations:

On any set S , $x \rho y \leftrightarrow x = y$.

On $\mathbb{N}$, $x \rho y \leftrightarrow x + y$ is even.

On the set of all lines in the plane, $x \rho y \leftrightarrow x$ is parallel to y or coincides with y .

On $\{0, 1\}$, $x \rho y \leftrightarrow x = y^2$.

On $\{x \mid x \text{ is a student in your class}\}$, $x \rho y \leftrightarrow x$ sits in the same row as y .

On $\{1, 2, 3\}$, $\rho = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$.

We can illustrate an important feature of an equivalence relation on a set by looking at $S = \{x \mid x \text{ is a student in your class}\}$, $x \rho y \leftrightarrow$ “ x sits in the same row as y .” Let’s group together all those students in set S who are related to one another. We come up with Figure 5.5. We have partitioned the set S into subsets in such a way that everyone in the class belongs to one and only one subset.

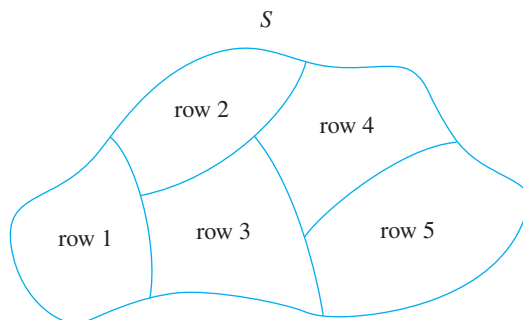


Figure 5.5

DEFINITION PARTITION OF A SET

A **partition** of a set S is a collection of nonempty disjoint subsets of S whose union equals S .

Any equivalence relation, as we will see, partitions the set on which it is defined. The subsets making up the partition, often called the **blocks** of the partition, are formed by grouping together related elements, like the students in the classroom.

For ρ an equivalence relation on a set S and $x \in S$, we let $[x]$ denote the set of all members of S to which x is related, called the **equivalence class** of x . Thus,

$$[x] = \{y \mid y \in S \wedge x \rho y\}$$

(Because ρ is symmetric, we could just as well have said that $[x] = \{y \mid y \in S \wedge y \rho x\}$.)

EXAMPLE 12

In the case where $x\rho y \leftrightarrow$ “ x sits in the same row as y ,” suppose that John, Chuck, Jose, Judy, and Ted all sit in row 3. Then $[\text{John}] = \{\text{John, Chuck, Jose, Judy, Ted}\}$. Also $[\text{John}] = [\text{Ted}] = [\text{Judy}]$, and so on. These are not distinct classes, but the same class with multiple names. An equivalence class can take its name from any of the elements in it. ●

Now we state the result about equivalence relations and partitions. For some practice with formal theorems and proofs, we give this result as a formal theorem, then analyze the structure of the proof and complete part of the proof.

● **THEOREM ON EQUIVALENCE RELATIONS AND PARTITIONS**

An equivalence relation ρ on a set S determines a partition of S , and a partition of a set S determines an equivalence relation on S .

Partial Proof: The theorem makes two separate statements:

- a. An equivalence relation on S determines a partition of S .
- b. A partition of S determines an equivalence relation on S .

To prove part (a), we must show that the distinct equivalence classes of members of S under equivalence relation ρ satisfy the definition of a partition. To satisfy the definition of a partition, we must show that

- i. the union of these distinct classes equals S .
- ii. the distinct classes are disjoint.

To prove part (a. i), we must show something about the union of the distinct equivalence classes formed by ρ . Equivalence classes are sets of elements of S , so their union is a set; let's denote this set by U . We must show that $U = S$, which is a set equality. To prove this set equality, we will prove set inclusion in each direction; in other words,

- 1. $U \subseteq S$
- 2. $S \subseteq U$

For this part of the proof, we are finally down to two small statements that are easy to prove, as follows:

a.i.1: Let $x \in U$. Then x belongs to an equivalence class. Every equivalence class is a subset of S , so $x \in S$.

a.i.2: Let $x \in S$. Then $x\rho x$ (reflexivity of ρ); thus, $x \in [x]$, and every member of S belongs to some equivalence class, hence to the union of classes U .

This completes the proof of part (a.i). For part (a.ii), let $[x]$ and $[z]$ be two equivalence classes. We want to show that distinct classes are disjoint, or

$$[x] \neq [z] \rightarrow [x] \cap [z] = \emptyset \quad (\text{a.ii})$$

If we assume that $[x] \neq [z]$, we must then show that $[x] \cap [z]$ does *not* contain anything, which might be hard to do. So instead, we'll prove the contrapositive of a. ii):

$$[x] \cap [z] \neq \emptyset \rightarrow [x] = [z] \quad (\text{contrapositive of a. ii})$$

Therefore, we assume that $[x] \cap [z] \neq \emptyset$ and that there is a $y \in S$ such that $y \in [x] \cap [z]$. What does this tell us?

$y \in [x] \cap [z]$	(assumption)
$y \in [x], y \in [z]$	(definition of $\cap$)
$x \rho y, z \rho y$	(definition of $[x]$ and $[z]$)
$x \rho y, y \rho z$	(symmetry of ρ)
$x \rho z$	(transitivity of ρ)

Now we can show that $[x] = [z]$ by proving set inclusion in each direction:

3. $[z] \subseteq [x]$
4. $[x] \subseteq [z]$

To show (3), $[z] \subseteq [x]$, let $q \in [z]$ (we know $[z] \neq \emptyset$ because $y \in [z]$.) Then

$z \rho q$	(definition of $[z]$)
$x \rho z$	(from above)
$x \rho q$	(transitivity of ρ)
$q \in [x]$	(definition of $[x]$)
$[z] \subseteq [x]$	(definition of $\subseteq$)

Practice 12 asks for a proof of (4), $[x] \subseteq [z]$. Once this proof is supplied, it completes (3) and (4), which leads to the conclusion $[x] = [z]$. This completes the proof of the contrapositive of part (a. ii) and therefore proves part (a. ii), which in turn completes the proof of part (a). Whew!

Practice 13 asks for a proof of part b.

End of Partial Proof

PRACTICE 12 For the foregoing argument, supply the proof that $[x] \subseteq [z]$. ■

PRACTICE 13 Prove part (b) of the theorem. Given a partition of a set S , define a relation ρ by

$$x \rho y \leftrightarrow x \text{ is in the same block of the partition as } y$$

and show that ρ is an equivalence relation on S , that is, show that ρ is reflexive, symmetric, and transitive. ■

EXAMPLE 13 The equivalence relation on $\mathbb{N}$ given by

$$x \rho y \leftrightarrow x + y \text{ is even}$$

partitions $\mathbb{N}$ into two equivalence classes. If x is an even number, then for any even number y , $x + y$ is even and $y \in [x]$. All even numbers form one class. If x is an odd number and y is any odd number, $x + y$ is even and $y \in [x]$. All odd numbers form the second class. The partition can be pictured as in Figure 5.6. Notice again that an equivalence class may have more than one name, or representative. In this example, $[2] = [8] = [1048]$, and so on; $[1] = [17] = [947]$, and so on.

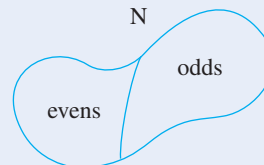


Figure 5.6

PRACTICE 14

For each of the following equivalence relations, describe the corresponding equivalence classes.

- On the set of all lines in the plane, $x\rho y \leftrightarrow x$ is parallel to y or x coincides with y .
- On the set $\mathbb{N}$, $x\rho y \leftrightarrow x = y$.
- On $\{1, 2, 3\}$, $\rho = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$.

Partitioning a set into equivalence classes is helpful because it is often convenient to go up one level of abstraction and treat the classes themselves as entities. We will conclude this section with two important examples where this is the case (you actually saw the first example somewhere around the fourth grade).

EXAMPLE 14

Let $S = \{a/b \mid a, b \in \mathbb{Z}, b \neq 0\}$. S is therefore the set of all fractions. The fractions $1/2$ and $2/4$ are different fractions—they have different numerators and different denominators, but they are said to be “equivalent.” Formally, a/b is equivalent to c/d , denoted by $a/b \sim c/d$, if and only if $ad = bc$. We will show that the binary relation $\sim$ on S is indeed an equivalence relation. First, $a/b \sim a/b$ because $ab = ba$. Also, if $a/b \sim c/d$ then $ad = bc$, or $cb = da$ and $c/d \sim a/b$. Hence, $\sim$ is reflexive and symmetric. To show that $\sim$ is transitive, let $a/b \sim c/d$ and $c/d \sim e/f$. Then $ad = bc$ and $cf = de$. Multiplying the first equation by f and the second by b , we get $adf = bcf$ and $bcf = bde$. Therefore, $adf = bde$, or $af = be$ (why is it legitimate to divide by d here?). Thus, $a/b \sim e/f$, and $\sim$ is transitive. Some sample equivalence classes of S formed by this equivalence relation are

$$\left[\frac{1}{2}\right] = \left\{\dots, \frac{-3}{-6}, \frac{-2}{-4}, \frac{-1}{-2}, \frac{1}{2}, \frac{2}{4}, \frac{3}{6}, \dots\right\}$$

$$\left[\frac{3}{10}\right] = \left\{\dots, \frac{-9}{-30}, \frac{-6}{-20}, \frac{-3}{-10}, \frac{3}{10}, \frac{6}{20}, \frac{9}{30}, \dots\right\}$$

The set $\mathbb{Q}$ of rational numbers can be regarded as the set of all equivalence classes of S . A single rational number, such as $[1/2]$, has many fractions representing it, although we customarily use the reduced fractional representation. When we add two rational numbers, such as $[1/2] + [3/10]$, we look for representatives

from the classes having the same denominator and add those representatives. Our answer is the class to which the resulting sum belongs, and we usually name the class by using a reduced fraction. Thus, to add $[1/2] + [3/10]$, we represent $[1/2]$ by $5/10$ and $[3/10]$ by $3/10$. The sum of $5/10$ and $3/10$ is $8/10$, and $[8/10]$ is customarily named $[4/5]$. This procedure is so familiar that it is generally written as $1/2 + 3/10 = 4/5$; nonetheless, equivalence classes of fractions are being manipulated by means of representatives. ●

EXAMPLE 15

We will define a binary relation of **congruence modulo 4** on the set $\mathbb{Z}$ of integers. An integer x is congruent modulo 4 to y , symbolized by $x \equiv_4 y$, or $x \equiv y \pmod{4}$, if $x - y$ is an integral multiple of 4. Congruence modulo 4 is an equivalence relation on $\mathbb{Z}$. To construct the equivalence classes, note that $[0]$, for example, will contain all integers differing from 0 by a multiple of 4, such as 4, 8, -12 , and so on. The distinct equivalence classes are

$$[0] = \{\dots, -8, -4, 0, 4, 8, \dots\}$$

$$[1] = \{\dots, -7, -3, 1, 5, 9, \dots\}$$

$$[2] = \{\dots, -6, -2, 2, 6, 10, \dots\}$$

$$[3] = \{\dots, -5, -1, 3, 7, 11, \dots\}$$

There is nothing special about the choice of 4 in Example 15; we can give a definition for **congruence modulo n** for any positive integer n .

● **DEFINITION CONGRUENCE MODULO n**

For integers x and y and positive integer n ,

$$x \equiv y \pmod{n} \text{ if } x - y \text{ is an integral multiple of } n$$

This binary relation is an equivalence relation on $\mathbb{Z}$ for any positive integer n (see Exercise 46). This equivalence relation and the resulting equivalence classes can be used for integer arithmetic on a computer. An integer is stored as a sequence of bits (0s and 1s) within a single memory location. Each computer allocates a fixed number of bits for a single memory location (this number varies depending on the architecture of the computer—how its memory space is laid out). The larger the integer, the more bits required to represent it. Therefore each machine has some limit on the size of the integers it can store. Suppose that $n - 1$ is the maximum size and that x and y are integer values with $0 \leq x \leq n - 1$, $0 \leq y \leq n - 1$. If the sum $x + y$ exceeds the maximum size, it cannot be stored. As an alternative, the computer may perform **addition modulo n** and find the remainder r when $x + y$ is divided by n .

The equation

$$x + y = qn + r, 0 \leq r < n$$

symbolizes this division, where q is the quotient and r is the remainder. This equation may be written as

$$(x + y) - r = qn$$

which shows that $(x + y) - r$ is an integral multiple of n , or that $(x + y) \equiv r \pmod{n}$. The integer r may not be $x + y$, but it is in the equivalence class $[x + y]$, and since $0 \leq r < n$, it is also in the range of integers that can be stored. (The system may or may not issue an *integer overflow* message if $x + y$ is too large to store and addition modulo n must be used.) The situation is analogous to your car's odometer, which records mileage modulo 100,000; when mileage reaches 102,758, for example, it is displayed on the odometer as 2,758.

PRACTICE 15 What are the equivalence classes corresponding to the relation of congruence modulo 5 on $\mathbb{Z}$?

PRACTICE 16 If 4 is the maximum integer that can be stored on a (micromicro) computer, what will be stored for the value $3 + 4$ if addition modulo 5 is used?

Table 5.1 summarizes important features of partial orderings and equivalence relations.

TABLE 5.1					
Partial Orderings and Equivalence Relations					
Type of Binary Relation	Reflexive	Symmetric	Antisymmetric	Transitive	Important Feature
Partial ordering	Yes	No	Yes	Yes	Predecessors and successors
Equivalence relation	Yes	Yes	No	Yes	Determines a partition

SECTION 5.1 REVIEW

TECHNIQUES

- Test an ordered pair for membership in a binary relation.
-  Test a binary relation for reflexivity, symmetry, antisymmetry, and transitivity.
- Find the reflexive, symmetric, and transitive closure of a relation.
- Draw the Hasse diagram for a finite partially ordered set.
- Find least, minimal, greatest, and maximal elements in a partially ordered set.
-  Find the equivalence classes associated with an equivalence relation.

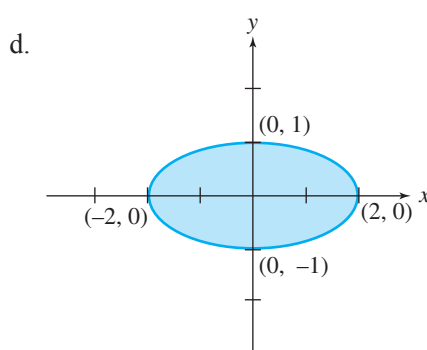
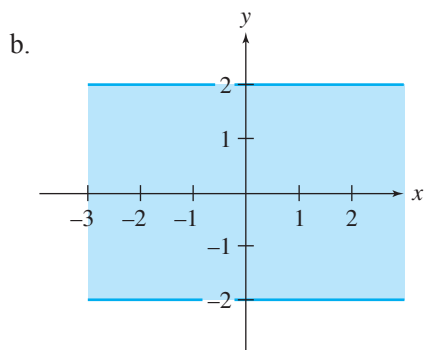
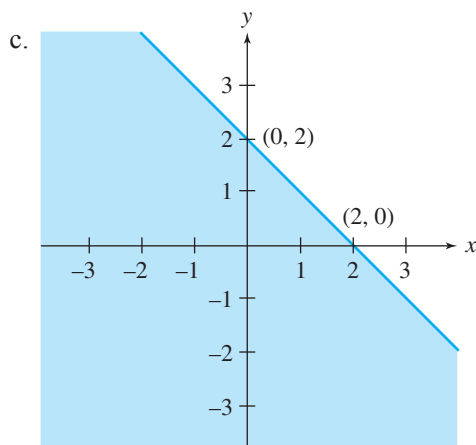
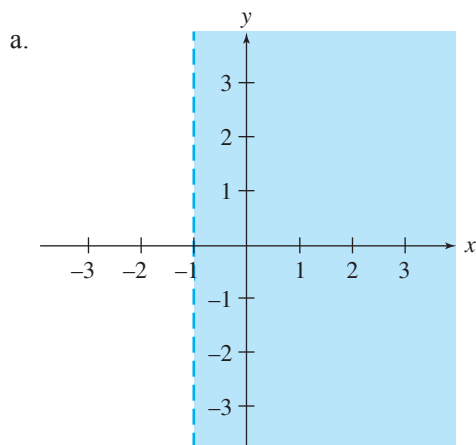
MAIN IDEAS

- A binary relation on a set S is formally a subset of $S \times S$; the distinctive relationship satisfied by the relation's members often has a verbal description as well.
- Operations on binary relations on a set include union, intersection, and complementation.
- Binary relations can have properties of reflexivity, symmetry, transitivity, and antisymmetry.
- Finite partially ordered sets can be represented graphically.
- An equivalence relation on a set S determines a partition of S , and conversely. The blocks of the partition are equivalence classes, which may themselves be treated as entities.

EXERCISES 5.1

- For each of the following binary relations ρ on $\mathbb{N}$, decide which of the given ordered pairs belong to ρ .
 - $x\rho y \leftrightarrow x + y < 7$; (1, 3), (2, 5), (3, 3), (4, 4)
 - $x\rho y \leftrightarrow x = y + 2$; (0, 2), (4, 2), (6, 3), (5, 3)
 - $x\rho y \leftrightarrow 2x + 3y = 10$; (5, 0), (2, 2), (3, 1), (1, 3)
 - $x\rho y \leftrightarrow y$ is a perfect square; (1, 1), (4, 2), (3, 9), (25, 5)
- For each of the following binary relations ρ on $\mathbb{Z}$, decide which of the given ordered pairs belong to ρ .
 - $x\rho y \leftrightarrow x|y$; (2, -6), (3, 5), (8, 4), (4, 8)
 - $x\rho y \leftrightarrow x$ and y are relatively prime; (5, 8), (9, 16), (6, 8), (8, 21)
 - $x\rho y \leftrightarrow \gcd(x, y) = 7$; (28, 14), (7, 7), (10, 5), (21, 14)
 - $x\rho y \leftrightarrow x^2 + y^2 = z^2$ for some integer z ; (1, 0), (3, 9), (2, 2), (-3, 4)
 - $x\rho y \leftrightarrow x$ is a number from the Fibonacci sequence; (4, 3), (7, 6), (7, 12), (20, 20)
- Decide which of the given items satisfy the relation.
 - ρ a binary relation on $\mathbb{Z}$, $x\rho y \leftrightarrow x = -y$; (1, -1), (2, 2), (-3, 3), (-4, -4)
 - ρ a binary relation on $\mathbb{N}$, $x\rho y \leftrightarrow x$ is prime; (19, 7), (21, 4), (33, 13), (41, 16)
 - ρ a binary relation on $\mathbb{Q}$, $x\rho y \leftrightarrow x \leq 1/y$; (1, 2), (-3, -5), (-4, 1/2), (1/2, 1/3)
 - ρ a binary relation on $\mathbb{N} \times \mathbb{N}$, $(x, y)\rho(u, v) \leftrightarrow x + u = y + v$; ((1, 2), (3, 2)), ((4, 5), (0, 1))
- Decide which of the given items satisfy the relation.
 - ρ a binary relation on $S \times C$, where $S = \{\text{states in the United States}\}$, $C = \{\text{cities in the United States}\}$, $x\rho y \leftrightarrow y$ is the capital of x ; (Indiana, Indianapolis), (Illinois, Chicago), (Kansas, Kansas City), (Kentucky, Louisville), (North Dakota, Bismarck)
 - ρ a binary relation on $A \times P$, where $A = \{\text{artists}\}$, $P = \{\text{paintings}\}$, $x\rho y \leftrightarrow x$ painted y ; (DaVinci, Mona Lisa), (Grant Wood, American Gothic), (Remington, Ridden Down), (Picasso, Blue Dancers), (van Gogh, Starry Night)
 - ρ a binary relation on $C \times M$, where $C = \{\text{composers}\}$, $M = \{\text{music}\}$, $x\rho y \leftrightarrow x$ composed y ; (Bernstein, West Side Story), (Presley, Blue Suede Shoes), (Gershwin, Rhapsody in Blue), (Beethoven, Moonlight Sonata), (Rogers and Hammerstein, Phantom of the Opera)
 - ρ a binary relation on $A \times B$, where $A = \{\text{authors}\}$, $B = \{\text{books}\}$, $x\rho y \leftrightarrow x$ wrote y ; (Hemingway, The Old Man and the Sea), (Sawyer, Huckleberry Finn) (Poe, Moby Dick), (Orwell, 1984), (Tolstoy, Crime and Punishment)
- For each of the following binary relations on $\mathbb{R}$, draw a figure to show the region of the plane it describes.
 - $x\rho y \leftrightarrow y \leq 2$
 - $x\rho y \leftrightarrow x = y - 1$
 - $x\rho y \leftrightarrow x^2 + y^2 \leq 25$
 - $x\rho y \leftrightarrow x \geq y$

6. For each of the accompanying figures, give the binary relation on $\mathbb{R}$ that describes the shaded area.



7. Identify each relation on $\mathbb{N}$ as one-to-one, one-to-many, many-to-one, or many-to-many.

- $\rho = \{(1, 2), (1, 4), (1, 6), (2, 3), (4, 3)\}$
- $\rho = \{(9, 7), (6, 5), (3, 6), (8, 5)\}$
- $\rho = \{(12, 5), (8, 4), (6, 3), (7, 12)\}$
- $\rho = \{(2, 7), (8, 4), (2, 5), (7, 6), (10, 1)\}$

8. Identify each of the following relations on S as one-to-one, one-to-many, many-to-one, or many-to-many.

- $S = \mathbb{N}$
 $x\rho y \leftrightarrow x = y + 1$
- $S = \text{set of all women in Vicksburg}$
 $x\rho y \leftrightarrow x \text{ is the daughter of } y$
- $S = \wp(\{1, 2, 3\})$
 $A\rho B \leftrightarrow |A| = |B|$
- $S = \mathbb{R}$
 $x\rho y \leftrightarrow x = 5$

9. Let ρ and σ be binary relations on $\mathbb{N}$ defined by $x\rho y \leftrightarrow$ “ x divides y ,” $x\sigma y \leftrightarrow 5x \leq y$. Decide which of the given ordered pairs satisfy the following relations.

- $\rho \cup \sigma; (2, 6), (3, 17), (2, 1), (0, 0)$
- $\rho \cap \sigma; (3, 6), (1, 2), (2, 12)$
- $\rho'; (1, 5), (2, 8), (3, 15)$
- $\sigma'; (1, 1), (2, 10), (4, 8)$

10. Both ρ and σ are binary relations from P to C where $P = \{\text{people in the United States}\}$, $C = \{\text{cities in the United States}\}$, $x\rho y \leftrightarrow x$ lives in y , and $x\sigma y \leftrightarrow x$ works in y . Describe each of the following relations.
- $\rho \cap \sigma$
 - $\rho \cup \sigma$
 - $\rho \cap \sigma'$
 - $\rho' \cap \sigma$
11. Let $S = \{1, 2, 3\}$. Test the following binary relations on S for reflexivity, symmetry, antisymmetry, and transitivity.
- $\rho = \{(1, 3), (3, 3), (3, 1), (2, 2), (2, 3), (1, 1), (1, 2)\}$
 - $\rho = \{(1, 1), (3, 3), (2, 2)\}$
 - $\rho = \{(1, 1), (1, 2), (2, 3), (3, 1), (1, 3)\}$
 - $\rho = \{(1, 1), (1, 2), (2, 3), (1, 3)\}$
12. Let $S = \{0, 1, 2, 4, 6\}$. Test the following binary relations on S for reflexivity, symmetry, antisymmetry, and transitivity.
- $\rho = \{(0, 0), (1, 1), (2, 2), (4, 4), (6, 6), (0, 1), (1, 2), (2, 4), (4, 6)\}$
 - $\rho = \{(0, 1), (1, 0), (2, 4), (4, 2), (4, 6), (6, 4)\}$
 - $\rho = \{(0, 1), (1, 2), (0, 2), (2, 0), (2, 1), (1, 0), (0, 0), (1, 1), (2, 2)\}$
 - $\rho = \{(0, 0), (1, 1), (2, 2), (4, 4), (6, 6), (4, 6), (6, 4)\}$
 - $\rho = \emptyset$
13. Test the following binary relations on the given sets S for reflexivity, symmetry, antisymmetry, and transitivity.
- $S = \mathbb{Q}$
 $x\rho y \leftrightarrow |x| \leq |y|$
 - $S = \mathbb{Z}$
 $x\rho y \leftrightarrow x - y$ is an integral multiple of 3
 - $S = \mathbb{N}$
 $x\rho y \leftrightarrow x \cdot y$ is even
 - $S = \mathbb{N}$
 $x\rho y \leftrightarrow x$ is odd
 - $S = \text{set of all squares in the plane}$
 $S_1\rho S_2 \leftrightarrow \text{length of side of } S_1 = \text{length of side of } S_2$
14. Test the following binary relations on the given sets S for reflexivity, symmetry, antisymmetry, and transitivity.
- $S = \text{set of all finite-length strings of characters}$
 $x\rho y \leftrightarrow \text{number of characters in } x = \text{number of characters in } y$
 - $S = \{0, 1, 2, 3, 4, 5\}$
 $x\rho y \leftrightarrow x + y = 5$
 - $S = \wp(\{1, 2, 3, 4, 5, 6, 7, 8, 9\})$
 $A\rho B \leftrightarrow |A| = |B|$
 - $S = \wp(\{1, 2, 3, 4, 5, 6, 7, 8, 9\})$
 $A\rho B \leftrightarrow |A| \neq |B|$
 - $S = \mathbb{N} \times \mathbb{N}$
 $(x_1, y_1)\rho(x_2, y_2) \leftrightarrow x_1 \leq x_2 \text{ and } y_1 \geq y_2$

15. Which of the binary relations of Exercise 13 are equivalence relations? For each equivalence relation, describe the associated equivalence classes.
16. Which of the binary relations of Exercise 14 are equivalence relations? For each equivalence relation, describe the associated equivalence classes.
17. Test the following binary relations on the given sets S for reflexivity, symmetry, antisymmetry, and transitivity.
 - a. $S = \mathbb{Z}$
 $x\rho y \leftrightarrow x + y$ is a multiple of 5
 - b. $S = \mathbb{Z}$
 $x\rho y \leftrightarrow x < y$
 - c. $S =$ set of all finite-length binary strings
 $x\rho y \leftrightarrow x$ is a prefix of y
 - d. $S =$ set of all finite-length binary strings
 $x\rho y \leftrightarrow x$ has the same number of 1s as y
18. Test the following binary relations on the given sets S for reflexivity, symmetry, antisymmetry, and transitivity.
 - a. $S = \mathbb{Z}$
 $x\rho y \leftrightarrow x = ky$ for some integer k
 - b. $S = \mathbb{Z}$
 $x\rho y \leftrightarrow$ there is a prime number p such that $p|x$ and $p|y$
 - c. $S = \wp(\{1, 2, 3, 4, 5, 6, 7, 8, 9\})$
 $A\rho B \leftrightarrow A \cap B = \emptyset$
 - d. $S = \wp(\{1, 2, 3, 4, 5, 6, 7, 8, 9\})$
 $A\rho B \leftrightarrow A = B'$
19. Let S be the set of people in the United States. Test the following binary relations on S for reflexivity, symmetry, antisymmetry, and transitivity.
 - a. $x\rho y \leftrightarrow x$ is at least as tall as y .
 - b. $x\rho y \leftrightarrow x$ is taller than y .
 - c. $x\rho y \leftrightarrow x$ is the same height as y .
 - d. $x\rho y \leftrightarrow x$ is a child of y .
20. Let S be the set of people in the United States. Test the following binary relations on S for reflexivity, symmetry, antisymmetry, and transitivity.
 - a. $x\rho y \leftrightarrow x$ is the husband of y .
 - b. $x\rho y \leftrightarrow x$ is the spouse of y .
 - c. $x\rho y \leftrightarrow x$ has the same parents as y .
 - d. $x\rho y \leftrightarrow x$ is the brother of y .
21. For each case, think of a set S and a binary relation ρ on S (different from any in the examples or problems) satisfying the given conditions.
 - a. ρ is reflexive and symmetric but not transitive.
 - b. ρ is reflexive and transitive but not symmetric.
 - c. ρ is not reflexive or symmetric but is transitive.
 - d. ρ is reflexive but neither symmetric nor transitive.

22. Let ρ and σ be binary relations on a set S .
- If ρ and σ are reflexive, is $\rho \cup \sigma$ reflexive? Is $\rho \cap \sigma$ reflexive?
 - If ρ and σ are symmetric, is $\rho \cup \sigma$ symmetric? Is $\rho \cap \sigma$ symmetric?
 - If ρ and σ are antisymmetric, is $\rho \cup \sigma$ antisymmetric? Is $\rho \cap \sigma$ antisymmetric?
 - If ρ and σ are transitive, is $\rho \cup \sigma$ transitive? Is $\rho \cap \sigma$ transitive?
23. Find the reflexive, symmetric, and transitive closure of each of the relations in Exercise 11.
24. Find the reflexive, symmetric, and transitive closure of each of the relations in Exercise 12.
25. Given the following binary relation

S = set of all cities in the country

$x\rho y \leftrightarrow$ Take-Your-Chance Airlines flies directly from x to y

describe in words what the transitive closure relation would be.

26. Two additional properties of a binary relation ρ are defined as follows:

ρ is *irreflexive* means: $(\forall x)(x \in S \rightarrow (x, x) \notin \rho)$

ρ is *asymmetric* means: $(\forall x)(\forall y)(x \in S \wedge y \in S \wedge (x, y) \in \rho \rightarrow (y, x) \notin \rho)$

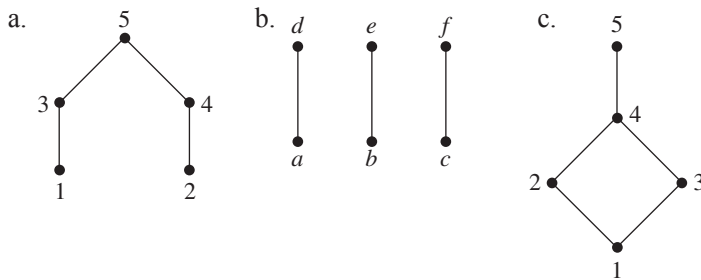
- Give an example of a binary relation ρ on set $S = \{1, 2, 3\}$ that is neither reflexive nor irreflexive.
 - Give an example of a binary relation ρ on set $S = \{1, 2, 3\}$ that is neither symmetric nor asymmetric.
 - Prove that if ρ is an asymmetric relation on a set S , then ρ is irreflexive.
 - Prove that if ρ is an irreflexive and transitive relation on a set S , then ρ is asymmetric.
 - Prove that if ρ is a nonempty, symmetric, and transitive relation on a set S , then ρ is not irreflexive.
27. Does it make sense to look for the irreflexive closure of a relation? (See Exercise 26.) Why or why not?
28. Does it make sense to look for the asymmetric closure of a relation? (See Exercise 26.) Why or why not?
29. Let S be an n -element set. How many different binary relations can be defined on S ? (*Hint*: Recall the formal definition of a binary relation.)
30. Let ρ be a binary relation on a set S . For $A \subseteq S$, define

$$\#A = \{x | x \in S \wedge (\forall y)(y \in A \rightarrow x\rho y)\}$$

$$A\# = \{x | x \in S \wedge (\forall y)(y \in A \rightarrow y\rho x)\}$$

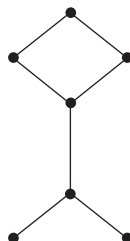
- Prove that if ρ is symmetric, then $\#A = A\#$.
 - Prove that if $A \subseteq B$ then $\#B \subseteq \#A$ and $B\# \subseteq A\#$.
 - Prove that $A \subseteq (\#A)\#$.
 - Prove that $A \subseteq \#(A\#)$.
31. Draw the Hasse diagram for the following partial orderings.
- $S = \{a, b, c\}$
 $\rho = \{(a, a), (b, b), (c, c), (a, b), (b, c), (a, c)\}$
 - $S = \{a, b, c, d\}$
 $\rho = \{(a, a), (b, b), (c, c), (d, d), (a, b), (a, c)\}$
 - $S = \{\emptyset, \{a\}, \{a, b\}, \{c\}, \{a, c\}, \{b\}\}$
 $A\rho B \leftrightarrow A \subseteq B$

32. For Exercise 31, name any least elements, minimal elements, greatest elements, and maximal elements.
33. Let $(S, \leq)$ be a partially ordered set, and let $A \subseteq S$. Prove that the restriction of $\leq$ to A is a partial ordering on A .
34. a. Draw the Hasse diagram for the partial ordering “ x divides y ” on the set $\{2, 3, 5, 7, 21, 42, 105, 210\}$. Name any least elements, minimal elements, greatest elements, and maximal elements. Name a totally ordered subset with four elements.
- b. Draw the Hasse diagram for the partial ordering “ x divides y ” on the set $\{3, 6, 9, 18, 54, 72, 108, 162\}$. Name any least elements, minimal elements, greatest elements, and maximal elements. Name any unrelated elements.
35. Draw the Hasse diagram for each of the two partially ordered sets.
- a. $S = \{1, 2, 3, 5, 6, 10, 15, 30\}$ b. $S = \emptyset(\{1, 2, 3\})$
 $x \rho y \leftrightarrow x \text{ divides } y$ $A \rho B \leftrightarrow A \subseteq B$
- What do you notice about the structure of these two diagrams?
36. For each Hasse diagram of a partial ordering in the accompanying figure, list the ordered pairs that belong to the relation.



37. Let (S, ρ) and (T, σ) be two partially ordered sets. A relation μ on $S \times T$ is defined by $(s_1, t_1) \mu (s_2, t_2) \leftrightarrow s_1 \rho s_2$ and $t_1 \sigma t_2$. Show that μ is a partial ordering on $S \times T$.
38. Let ρ be a binary relation on a set S . Then a binary relation called the inverse of ρ , denoted by ρ^{-1} , is defined by $x \rho^{-1} y \leftrightarrow y \rho x$.
- a. For $\rho = \{(1, 2), (2, 3), (5, 3), (4, 5)\}$ on the set $\mathbb{N}$, what is ρ^{-1} ?
- b. Prove that if ρ is a reflexive relation on a set S , then ρ^{-1} is reflexive.
- c. Prove that if ρ is a symmetric relation on a set S , then ρ^{-1} is symmetric.
- d. Prove that if ρ is an antisymmetric relation on a set S , then ρ^{-1} is antisymmetric.
- e. Prove that if ρ is a transitive relation on a set S , then ρ^{-1} is transitive.
- f. Prove that if ρ is an irreflexive relation on a set S (see Exercise 26), then ρ^{-1} is irreflexive.
- g. Prove that if ρ is an asymmetric relation on a set S (see Exercise 26), then ρ^{-1} is asymmetric.
39. Prove that if a binary relation ρ on a set S is reflexive and transitive, then the relation $\rho \cap \rho^{-1}$ is an equivalence relation (see Exercise 38 for the definition of ρ^{-1}).
40. a. Let (S, ρ) be a partially ordered set. Then ρ^{-1} can be defined as in Exercise 38. Show that (S, ρ^{-1}) is a partially ordered set, called the *dual* of (S, ρ) .

- b. If (S, ρ) is a finite, partially ordered set with the Hasse diagram shown, draw the diagram of the dual of (S, ρ) .



- c. Let (S, ρ) be a totally ordered set and let $X = \{(x, x) | x \in S\}$. Show that the set difference $\rho^{-1} - X$ equals the set ρ' .
41. A computer program is to be written that will generate a dictionary or the index for a book. We will assume a maximum length of n characters per word. Thus, we are given a set S of words of length at most n , and we want to produce a linear list of these words arranged in alphabetical order. There is a natural total ordering $\leq$ on alphabetic characters ($a < b$, $b < c$, etc.), and we will assume our words contain only alphabetic characters. We want to define a total ordering $\leq$ on S called a *lexicographical ordering* that will arrange the members of S alphabetically. The idea is to compare two words X and Y character by character, passing over equal characters. If at any point the X character alphabetically precedes the corresponding Y character, then X precedes Y ; if all characters in X are equal to the corresponding Y characters but we run out of characters in X before characters in Y , then X precedes Y . Otherwise, Y precedes X .

Formally, let $X = (x_1, x_2, \dots, x_j)$ and $Y = (y_1, y_2, \dots, y_k)$ be members of S with $j \leq k$. Let β (for blank) be a new symbol, and fill out X with $k - j$ blanks on the right. X can now be written $(x_1, x_2, \dots, x_k)$. Let β precede any alphabetical character. Then $X \leq Y$ if

$$x_1 \neq y_1 \text{ and } x_1 \leq y_1$$

or

$$\begin{aligned} x_1 = y_1, x_2 = y_2, \dots, x_m = y_m (m \leq k) \\ x_{m+1} \neq y_{m+1} \text{ and } x_{m+1} \leq y_{m+1} \end{aligned}$$

Otherwise, $Y \leq X$.

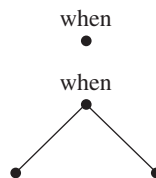
Note that because the ordering $\leq$ on alphabetical characters is a total ordering, if $Y \leq X$ by “otherwise,” then there exists $m \leq k$ such that $x_1 = y_1, x_2 = y_2, \dots, x_m = y_m, x_{m+1} \neq y_{m+1}$ and $y_{m+1} \leq x_{m+1}$.

Show that $\leq$ on S as defined above is a total ordering.

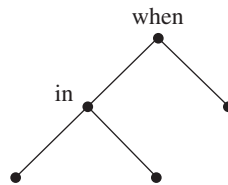
42. Apply the total ordering described in Exercise 41 to the words *boo*, *bug*, *be*, *bah*, and *bugg*. Note why each word precedes the next.
43. Exercise 41 discusses a total ordering on a set of words of length at most n that will produce a linear list in alphabetical order. Suppose we want to generate a list of all the distinct words in a text (for example, a compiler must create a symbol table of variable names). As in Exercise 41, we assume that the words contain only alphabetical characters because there is a natural precedence relation already existing ($a < b$, $b < c$, and so on). If numeric or special characters are involved, they must be assigned a

precedence relation with alphabetical characters (the collating sequence must be determined). If we list words alphabetically, it is a fairly quick procedure to decide whether a word currently being processed is new, but to fit the new word into place, all successive words must be moved one unit down the line. If the words are listed in the order in which they are processed, new words are simply tacked onto the end and no rearranging is necessary, but each word being processed has to be compared with each member of the list to determine if it is new. Thus, both logical linear lists have disadvantages.

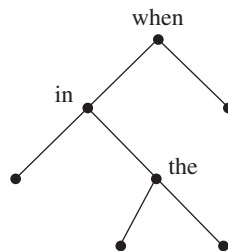
We describe a structure called a *binary search tree*; using this structure, a search process called a *binary tree search* can usually determine quickly whether a word is new and, if it is, no juggling is required to fit it into place, thus eliminating the disadvantages of both linear list structures described earlier. Suppose we want to process the phrase “when in the course of human events.” The first word in the text is used to label the first node of a graph. Once a node is labeled, it drops down a left and right arc, putting two unlabeled nodes below the one just labeled.



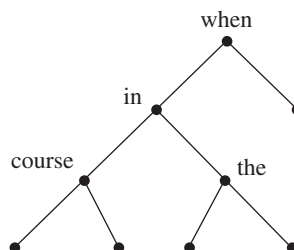
When the next word in the text is processed, it is compared with the first node. When the word being processed alphabetically precedes the label of a node, the left arc is taken; when the word follows the label alphabetically, the right arc is taken. The word becomes the label of the first unlabeled node it reaches. (If the word equals a node label, it is a duplicate, so the next word in the text is processed.) This procedure continues for the entire text. Thus,



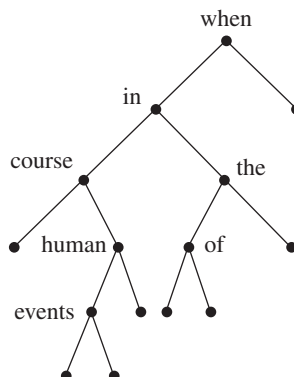
then



then



until finally



By traversing the nodes of this graph in the proper order (described by always processing the left nodes below a node first, then the node, then the right nodes below it), an alphabetical listing “course, events, human, in, of, the, when” is produced.

- a. This type of graph is called a *tree*. (Unlabeled nodes and arcs to unlabeled nodes usually are not shown.) Turned upside down, the graph can be viewed as the Hasse diagram of a partial ordering $\leq$. What would be the least element? Would there be a greatest element? Which of the following ordered pairs would belong to $\leq$: (in, of), (the, of), (in, events), (course, of)?

Here the tree structure contains more information than the partial ordering, because we are interested in not only whether a word w_1 precedes a word w_2 in the partial ordering sense but also whether w_2 is to the left or right of w_1 .

- b. Build a binary search tree for the phrase “Old King Cole was a merry old soul.” Eliminate unlabeled nodes. Considering the upside-down graph as the Hasse diagram of a partial ordering, name the maximal elements.
44. The alphabetical ordering defined in Exercise 41 can be applied to words of any finite length. If we define A^* to be the set of all finite-length “words” (strings of characters, not necessarily meaningful) from the English alphabet, then the alphabetical ordering on A^* has all words composed only of the letter A preceding all other words. Thus, all the words in the infinite list

$a, aa, aaa, aaaa, \dots$

precede words such as “ b ” or “ $aaaaaab$.” Therefore this list does not enumerate A^* , because we can never count up to any words with any characters other than a . However, the set A^* is denumerable. Write a partial enumeration of A^* by ordering words by length (all words of length 1 precede all words of length 2, and so on) and then alphabetically ordering words of the same length.

45. a. For the equivalence relation $\rho = \{(a, a), (b, b), (c, c), (a, c), (c, a)\}$, what is the set $[a]$? Does it have any other names?
- b. For the equivalence relation $\rho = \{(1, 1), (2, 2), (1, 2), (2, 1), (1, 3), (3, 1), (3, 2), (2, 3), (3, 3), (4, 4), (5, 5), (4, 5), (5, 4)\}$ what is the set $[3]$? What is the set $[4]$?
46. Prove that for any positive integer n , congruence modulo n is an equivalence relation on the set $\mathbb{Z}$.
47. For the equivalence relation of congruence modulo 2 on the set $\mathbb{Z}$, what is the set $[1]$?
48. For the equivalence relation of congruence modulo 5 on the set $\mathbb{Z}$, what is the set $[-3]$?
49. Assume that $x \equiv y \pmod{n}$ and $z \equiv w \pmod{n}$ for some positive integer n . Prove that
- $x + z \equiv y + w \pmod{n}$
 - $x - z \equiv y - w \pmod{n}$
 - $x \cdot z \equiv y \cdot w \pmod{n}$
 - $x^s \equiv y^s \pmod{n}$ for $s \geq 1, n \geq 2$

50. Let p be a prime number. Prove that $x^2 \equiv y^2 \pmod{p}$ if and only if $x \equiv y \pmod{p}$ or $x \equiv -y \pmod{p}$.
51. a. Given the partition $\{1, 2\}$ and $\{3, 4\}$ of the set $S = \{1, 2, 3, 4\}$, list the ordered pairs in the corresponding equivalence relation.
 b. Given the partition $\{a, b, c\}$ and $\{d, e\}$ of the set $S = \{a, b, c, d, e\}$, list the ordered pairs in the corresponding equivalence relation.
52. Let S be the set of all books in the library. Let ρ be a binary relation on S defined by $x\rho y \leftrightarrow$ “the color of x ’s cover is the same as the color of y ’s cover.” Show that ρ is an equivalence relation on S and describe the resulting equivalence classes.
53. Let $S = \mathbb{N}$ and let ρ be a binary relation on S defined by $x\rho y \leftrightarrow x^2 - y^2$ is even. Show that ρ is an equivalence relation on S and describe the resulting equivalence classes.
54. Let $S = \mathbb{R}$ and let ρ be a binary relation on S defined by $x\rho y \leftrightarrow x - y$ is an integer.
 a. Show that ρ is an equivalence relation on S .
 b. List 5 values that belong to $[1.5]$.
55. Let $S = \mathbb{N} \times \mathbb{N}$ and let ρ be a binary relation on S defined by $(x, y)\rho(z, w) \leftrightarrow y = w$. Show that ρ is an equivalence relation on S and describe the resulting equivalence classes.
56. Let $S = \mathbb{N} \times \mathbb{N}$ and let ρ be a binary relation on S defined by $(x, y)\rho(z, w) \leftrightarrow x + y = z + w$. Show that ρ is an equivalence relation on S and describe the resulting equivalence classes.
57. Let S be the set of all binary strings of length 8 and let ρ be a binary relation on S defined by $x\rho y \leftrightarrow$ y starts with the same bit value (0 or 1) as x and y ends with the same bit value (0 or 1) as x .
 a. Show that ρ is an equivalence relation on S .
 b. How many strings are in the set S ?
 c. Into how many equivalence classes does ρ partition S ? Explain your answer.
 d. How many strings are in each equivalence class?
58. The documentation for the Java programming language recommends that when a boolean “equals method” is defined for an object, it should be an equivalence relation. That is, if ρ is defined by $x\rho y \leftrightarrow x.\text{equals}(y)$ for all objects in the class, then ρ should be an equivalence relation. In a graphics application, a programmer creates an object called a point, consisting of two coordinates in the plane. The programmer defines an equals method as follows: If p and q are any two points in the plane, then

$$p.\text{equals}(q) \leftrightarrow \text{the distance from } p \text{ to } q \text{ is } \leq c$$

where c is a small positive number that depends on the resolution of the computer display. Is the programmer’s equals method an equivalence relation? Justify your answer.

59. Let S be the set of all propositional wffs with n statement letters. Let ρ be a binary relation on S defined by $P\rho Q \leftrightarrow$ “ $P \leftrightarrow Q$ is a tautology.” Show that ρ is an equivalence relation on S and describe the resulting equivalence classes. (We have used the notation $P \Leftrightarrow Q$ for $P\rho Q$.)
60. Given two partitions π_1 and π_2 of a set S , π_1 is a *refinement* of π_2 if each block of π_1 is a subset of a block of π_2 . Show that refinement is a partial ordering on the set of all partitions of S .

Exercises 61–72 all deal with partitions on a set.

61. Let P_n denote the total number of partitions of an n -element set, $n \geq 1$. The numbers P_n are called Bell numbers. Compute the following Bell numbers.
 a. P_1
 b. P_2
 c. P_3
 d. P_4

62. From Exercise 61, you might be looking for a closed-form formula giving the value of P_n . Although Bell numbers have been extensively studied, no closed-form formula has been found. Bell numbers can be computed via a recurrence relation. Let P_0 have the value 1. Prove that for $n \geq 1$,

$$P_n = \sum_{k=0}^{n-1} C(n-1, k)P_k$$

(Hint: Use a combinatorial proof instead of an inductive proof. Let x be a fixed but arbitrary member of a set with n elements. In each term of the sum, $n - k$ represents the size of the partition block that contains x .)

63. Use the formula of Exercise 62 to compute P_1 , P_2 , P_3 , and P_4 , and compare your answers to those in Exercise 61.
64. Use the formula of Exercise 62 to compute P_5 and P_6 .
65. Let $S(n, k)$ denote the number of ways to partition a set of n elements into k blocks. The numbers $S(n, k)$ are called *Stirling numbers*.
- Find $S(3, 2)$.
 - Find $S(4, 2)$.
66. Prove that for all $n \geq 1$, $S(n, k)$ satisfies the recurrence relation

$$S(n, 1) = 1$$

$$S(n, n) = 1$$

$$S(n+1, k+1) = S(n, k) + (k+1)S(n, k+1) \text{ for } 1 \leq k \leq n$$

(Hint: Use a combinatorial proof instead of an inductive proof. Let x be a fixed but arbitrary member of a set with $n+1$ elements, and put x aside. Partition the remaining set of n elements. A partition of the original set could be obtained either by adding $\{x\}$ as a separate block or by putting x in one of the existing blocks.)

67. Use the formula of Exercise 66 to rework Exercise 65.
68. The recurrence relation of Exercise 66 is similar to Pascal's formula, Equation (1) of Section 4.5. Use this relation to compute the numeric values in the first five rows of *Stirling's triangle*, which begins

$$\begin{array}{c} S(1, 1) \\ S(2, 1) \quad S(2, 2) \\ S(3, 1) \quad S(3, 2) \quad S(3, 3) \\ \vdots \end{array}$$

69. Prove that

$$P_n = \sum_{k=1}^n S(n, k)$$

70. Use the formula of Exercise 69 and Stirling's triangle (Exercise 68) to compute P_1 , P_2 , P_3 , and P_4 .
71. Find the number of ways to distribute 4 different-colored marbles among 3 identical containers so that no container is empty.
72. Find the number of ways in which 5 different jobs can be assigned to 3 identical processors so that each processor gets at least 1 job.

73. Binary relations on a set S are ordered pairs of elements of S . More generally, an n -ary relation on a set S is a set of ordered n -tuples of elements of S . Decide which of the given items satisfy the relation.

a. ρ a unary relation on $\mathbb{Z}$, $x \in \rho \leftrightarrow x$ is a perfect square

25, 39, 49, 62

b. ρ a ternary relation on $\mathbb{N}$, $(x, y, z) \in \rho \leftrightarrow x^2 + y^2 = z^2$

(1, 1, 2), (3, 4, 5), (0, 5, 5), (8, 6, 10)

c. ρ a 4-ary relation on $\mathbb{Z}$, $(x, y, z, w) \in \rho \leftrightarrow y = |x|$ and $w \geq x + z^2$

(-4, 4, 2, 0), (5, 5, 1, 5), (6, -6, 6, 45), (-6, 6, 0, -2)

74. A ternary relation ρ is defined on the set $S = (2, 4, 6, 8)$ by $(x, y, z) \in \rho \leftrightarrow x + y = z$. List the 3-tuples that belong to ρ .

75. If x is a real number, $x \neq 0$, then a number y such that $x \cdot y = 1$ is called the multiplicative inverse of x . Given positive integers x and n , a positive integer y such that $x \cdot y \equiv 1 \pmod{n}$ is called the *modular multiplicative inverse of x modulo n* . But

$$x \cdot y \equiv 1 \pmod{n}$$

$$\leftrightarrow x \cdot y - 1 = kn \text{ where } k \text{ is an integer}$$

$$\leftrightarrow xy - kn = 1$$

$$\leftrightarrow 1 \text{ is a linear combination of } x \text{ and } n$$

$$\leftrightarrow \gcd(x, n) = 1$$

$$\leftrightarrow x \text{ and } n \text{ are relatively prime}$$

Thus, if x and n are not relatively prime, the modular inverse of x does not exist. If they are relatively prime, the modular inverse of x is the positive coefficient of x in the linear combination of x and n that equals 1.

Use the Euclidean algorithm to find the modular multiplicative inverse of 21 modulo 25 (note that 21 and 25 are relatively prime).

76. Use the Euclidean algorithm to find the modular multiplicative inverse of 68 modulo 15 (see Exercise 75).

SECTION 5.2 TOPOLOGICAL SORTING

If ρ is a partial ordering on a set S , then some elements of S are predecessors of other elements. If S is a set of tasks that are to be done, then the idea of x as a predecessor of y can be interpreted literally to mean that task x must be done before task y . Thus partial orderings and Hasse diagrams are natural ways to represent problems in task scheduling.

EXAMPLE 16

Ernie and his brothers run a woodworking shop in the hills of New Hampshire that manufactures rocking chairs with padded cushion seats. The manufacturing process can be broken down into a number of tasks, some of which have certain other tasks as prerequisites. The following table shows the manufacturing tasks for a rocking chair, the prerequisite tasks, and the number of hours required to perform each task.

Task	Prerequisite Tasks	Hours to Perform
1. Selecting wood	None	3.0
2. Carving rockers	1	4.0
3. Carving seat	1	6.0
4. Carving back	1	7.0
5. Carving arms	1	3.0
6. Selecting fabric	None	1.0
7. Sewing cushion	6	2.0
8. Assembling back and seat	3, 4	2.0
9. Attaching arms	5, 8	2.0
10. Attaching rockers	2, 8	3.0
11. Varnishing	9, 10	5.0
12. Adding cushion	7, 11	0.5

We can define a partial ordering on the set of tasks by

$$x \leq y \leftrightarrow \text{task } x = \text{task } y \text{ or task } x \text{ is a prerequisite to task } y$$

It is easy to see that this relation is reflexive, antisymmetric, and transitive. Also,

$$x < y \leftrightarrow \text{task } x \text{ is a prerequisite to task } y$$

In the Hasse diagram for this partial ordering, the nodes are tasks; we'll add to each node the information about the time to perform the task. Also, as is traditional, we'll orient the diagram so that if $x < y$, then x is to the left of y rather than below y . Thus the entire diagram runs from left to right rather than from bottom to top. Such a diagram for task scheduling is often called a **PERT(program evaluation and review technique)** chart, first developed for tracking the construction of Navy submarines but useful for managing any complex project with a number of subtasks. The PERT chart for manufacturing rocking chairs is shown in Figure 5.7, with task numbers substituted for task names and arrows pointing to a task from its prerequisite task(s). The numbers in parentheses indicate the time required to perform the task.

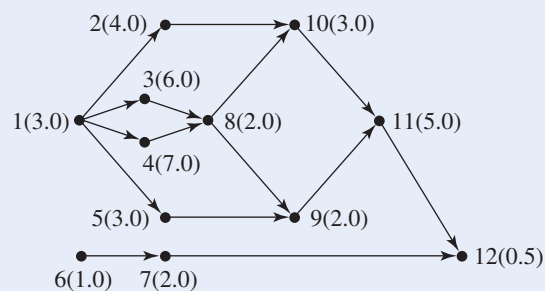


Figure 5.7

PRACTICE 17 Construct the PERT chart for building a house from the following task table.

Task	Prerequisite Tasks	Days to Perform
1. Clearing lot	None	4
2. Pouring pad	1	3
3. Doing framing	2	7
4. Shingling roof	3	6
5. Adding outside siding	3	4
6. Installing plumbing and wiring	4, 5	6
7. Hanging windows and doors	3	5
8. Installing wallboard	6	5
9. Painting interior	7, 8	5

A project represented by a PERT chart must begin with the tasks at the leftmost edge of the PERT chart and end with the tasks at the rightmost edge. An upper limit on the time required to complete the project can be obtained by adding the times for performing each task, but this does not take into account the fact that perhaps some tasks can be performed in parallel, such as tasks 2 through 5 in Example 16. To obtain the minimum time required to complete the project, we can move through the chart from left to right, computing for each node the minimum time to complete the work from the beginning through the work at that node. If a node x has multiple nodes as prerequisites, all the prerequisite tasks must be completed before we can begin work on x ; thus we must add the time for task x to the maximum completion time of the prerequisite nodes.

EXAMPLE 17 Let's compute the time for completing each task in Example 16.

Task 1: 3.0
 Task 2: $3.0 + 4.0 = 7.0$
 Task 3: $3.0 + 6.0 = 9.0$
 Task 4: $3.0 + 7.0 = 10.0$
 Task 5: $3.0 + 3.0 = 6.0$
 Task 6: 1.0
 Task 7: $1.0 + 2.0 = 3.0$
 Task 8: $\max(\text{time to complete task 3, time to complete task 4})$
 $+ \text{time to perform task 8}$
 $= \max(9.0, 10.0) + 2.0 = 10.0 + 2.0 = 12.0$

Task 9:	$\max(\text{time to complete task 5, time to complete task 8})$ $+ \text{time to perform task 9}$ $= \max(6.0, 12.0) + 2.0 = 12.0 + 2.0 = 14.0$
Task 10:	$\max(\text{time to complete task 2, time to complete task 8})$ $+ \text{time to perform task 10}$ $= \max(7.0, 12.0) + 3.0 = 12.0 + 3.0 = 15.0$
Task 11:	$\max(\text{time to complete task 9, time to complete task 10})$ $+ \text{time to perform task 11}$ $= \max(14.0, 15.0) + 5.0 = 15.0 + 5.0 = 20.0$
Task 12:	$\max(\text{time to complete task 7, time to complete task 11})$ $+ \text{time to perform task 12}$ $= \max(3.0, 20.0) + 0.5 = 20.0 + 0.5 = 20.5$

Therefore the minimum number of hours to manufacture a rocking chair is 20.5. From node 12, we can travel back in the chart, selecting at each point of multiple prerequisites the node that contributed the maximum value. This gives the sequence of nodes

12, 11, 10, 8, 4, 1

or, reversing this sequence,

1, 4, 8, 10, 11, 12

The sum of the times to perform each task in this sequence is 20.5. If any of these tasks takes longer to perform than its allotted time, the entire project will take longer than 20.5 hours. This sequence of nodes is a **critical path** through the PERT chart—performing these tasks in the allotted time is critical to completing the entire project on time. ●

The critical path in a PERT chart represents the minimum time to completion of the entire project. If a task not on the critical path takes longer than its allotted time to perform, then the critical path may shift to include this node, because it then becomes the bottleneck slowing down completion of the total project. In a complex project, the critical path must continually be recomputed to determine where best to allocate resources to move the project forward.

PRACTICE 18

Compute the minimum time to completion and the nodes on the critical path for the house-building project of Practice 17. ■

Given a partial ordering ρ on a finite set, there is always a total ordering σ that is an extension of ρ , meaning that if $x\rho y$, then $x\sigma y$. The process of **topological sorting** finds such a total ordering from a partial ordering. This is indeed a sorting process in the sense that the objects end up being totally ordered, but since they must be partially ordered to begin with, it is a very specialized sorting process.

Recall that in a finite partially ordered set, an element is minimal if it has no predecessors. In a finite nonempty partially ordered set, at least one minimal

element must exist. To see this, let x belong to the set. If x is not minimal, then there is a y in the set with $y \rho x$, $y \neq x$. If y is not minimal, then there is a z in the set with $z \rho y$, $z \neq y$, and so on. Because the set is finite, this process cannot go on indefinitely, so one such element must be minimal. A minimal element in a Hasse diagram has no elements below it; a minimal element in a PERT chart has no elements to its left.

The accompanying pseudocode algorithm for topological sorting operates on a partially ordered set (S, ρ) . Minimal elements (picked at random if there is a choice of minimal elements at any stage) are repeatedly removed from the ordered set until the set is empty. Each removal of a minimal element leaves a finite partially ordered set, so that another minimal element may be found.

ALGORITHM *TOPOLOGICALSORT*

```

TopSort(finite set  $S$ ; partial ordering  $\rho$  on  $S$ )
//find a total ordering on  $S$  that is an extension of  $\rho$ 
Local variable
integer  $i$       //enumerates tasks in total ordering
 $i = 1$ 
while  $S \neq \emptyset$ 
    pick a minimal element  $x_i$  from  $S$ ;
     $S = S - \{x_i\}$ 
     $i = i + 1$ 
end while
// $x_1 < x_2 < x_3 < \dots < x_n$  is now a total ordering that extends  $\rho$ 
write( $x_1, x_2, x_3, \dots, x_n$ )
end function TopSort

```

The ordering $x_1 < x_2 < x_3 < \dots < x_n$ produced by this algorithm is a total ordering. To see that it is an extension of ρ , suppose that $x_i \rho x_j$. Then x_i precedes x_j and x_i must be chosen as a minimal element and removed from the set before x_j can be chosen as a minimal element. Therefore $i < j$ and $x_i < x_j$.

EXAMPLE 18

One topological sort of the partial ordering of Example 16 is

6, 1, 7, 2, 3, 5, 4, 8, 10, 9, 11, 12

In Figure 5.7, either 6 or 1 is minimal and may be chosen as the first element. If 6 is chosen and removed from the set, then, as shown in Figure 5.8, either 1 or 7 is minimal. If 1 is then chosen and removed from the set (Figure 5.9), then 2, 3, 4, 5, and 7 are all minimal and any one can be chosen next. The process continues until all nodes have been chosen. If Ernie's brothers all move to the city and he is left to build rocking chairs alone, the topological sort gives an order in which he can perform tasks sequentially.

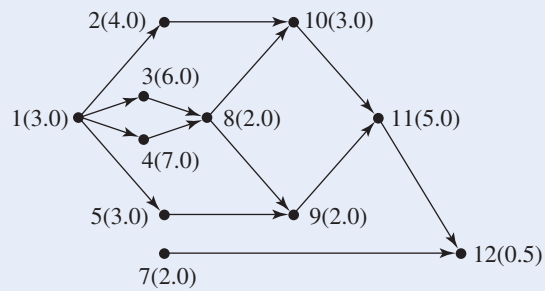


Figure 5.8

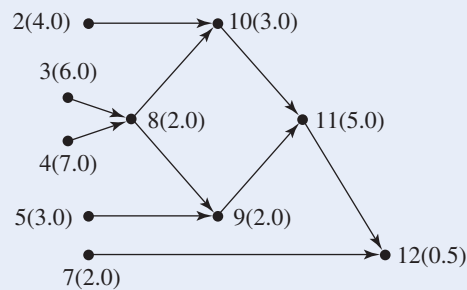


Figure 5.9

PRACTICE 19 Find another topological sort for the partial ordering of Example 16.

PRACTICE 20 Find a topological sort for the partial ordering of Practice 17.

The algorithm given here for topological sorting is still somewhat imprecise, as we have not given a mechanical method for finding a minimal element. Another algorithm will be described in Section 7.4.

SECTION 5.2 REVIEW

TECHNIQUES

- W Construct a PERT chart from a task table.
 - Find the critical path in a PERT chart.
- W Do a topological sort on a partially ordered set.

MAIN IDEAS

- PERT charts are diagrams of partially ordered sets representing tasks and prerequisites among tasks.
- A topological sort extends a partial ordering on a finite set to a total ordering.

EXERCISES 5.2

1. The following tasks are required in order to assemble a bicycle. As the manufacturer, you must write a list of sequential instructions for the buyer to follow. Will the sequential order given below work? Give another sequence that could be used.

Task	Prerequisite Tasks
1. Tightening frame fittings	None
2. Attaching handle bars to frame	1
3. Attaching gear mechanism	1
4. Mounting tire on wheel assembly	None
5. Attaching wheel assembly to frame	1, 4
6. Installing brake mechanism	2, 3, 5
7. Adding pedals	6
8. Attaching seat	1
9. Adjusting seat height	7, 8

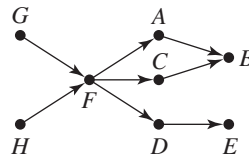
2. Give a list of tasks and prerequisite tasks for cooking and serving a hamburger.
 3. Construct a PERT chart from the following task table.

Task	Prerequisite Tasks	Time to Perform
<i>A</i>	<i>E</i>	3
<i>B</i>	<i>C, D</i>	5
<i>C</i>	<i>A</i>	2
<i>D</i>	<i>A</i>	6
<i>E</i>	None	2
<i>F</i>	<i>A, G</i>	4
<i>G</i>	<i>E</i>	4
<i>H</i>	<i>B, F</i>	1

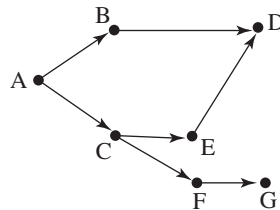
4. Construct a PERT chart from the following task table.

Task	Prerequisite Tasks	Time to Perform
1	2	4
2	3	2
3	8	5
4	3	2
5	4, 7	2
6	5	1
7	3	3
8	None	5

5. Compute the minimum time to completion and the nodes on the critical path for the problem in Exercise 3.
6. Compute the minimum time to completion and the nodes on the critical path for the problem in Exercise 4.
7. For the problem in Exercise 3, great improvements in productivity have knocked down the time to perform task *D* from 6 units to 1 unit. Recompute the minimum time to completion and the nodes on the critical path.
8. For the problem in Exercise 4, an extra quality-control step has been added to task 4, which now requires 4 units of time to perform. Recompute the minimum time to completion and the nodes on the critical path.
9. Do a topological sort on the partially ordered set shown.



10. Do a topological sort on the partially ordered set shown.



11. Find a topological sort for the problem in Exercise 3.
12. Find a topological sort for the problem in Exercise 4.
13. Given the following task chart for fixing an Asian dinner, find a total ordering in which the tasks can be performed sequentially.

Task	Prerequisite Tasks
1. Chop onions	9
2. Wash lettuce	11
3. Make dressing	11
4. Do stir fry	10
5. Toss salad	2, 3
6. Cut up chicken	None
7. Grate ginger	9
8. Chop bok choy	9
9. Marinate chicken	6
10. Heat wok	1, 7, 8, 11
11. Prepare rice	None

14. A U.S. journalist, on being posted to a bureaucratic foreign country, was faced with the following tasks before she could begin work.

Task	Prerequisite Tasks
1. Obtain a residence permit from the Public Security Bureau	2, 3, 7
2. Obtain a health certificate from the local hospital	None
3. Obtain a journalist work card from the Foreign Ministry	None
4. Obtain a customs certificate from the Customs Office	1, 3, 9
5. Post an announcement in the local newspaper about the presence of her news company in the country	None
6. Obtain a journalist visa from the Public Security Bureau	2, 3, 7
7. Obtain a foreign journalist housing contract from the local housing authority	None
8. Pick up her shipment of belongings from the United States	1, 4, 6
9. Obtain a news organization permit from the Foreign Ministry.	5

Find a total ordering in which the tasks can be performed sequentially.

15. Recall the problem posed at the beginning of this chapter:

Your company has developed a program for use on a small parallel processing machine. According to the technical documentation, the program executes processes P1, P2, and P3 in parallel; these processes all need results from process P4, so they must wait for Process P4 to complete execution before they begin. Processes P7 and P10 execute in parallel but must wait until Processes P1, P2, and P3 have finished. Process P4 requires results from P5 and P6 before it can begin execution. P5 and P6 execute in parallel. Processes P8 and P11 execute in parallel but P8 must wait for Process P7 to complete, and Process P11 must wait for P10 to complete. Process P9 must wait for results from P8 and P11. You have been assigned to convert the software for use on a single processor machine.

Use a topological sort to determine the order in which the processes should be executed sequentially.

16. Given the following task chart for conducting a water quality study, find a total ordering in which the tasks can be performed sequentially.

Task	Prerequisite Tasks
1. Plan schedule	7
2. Collect new data	1, 10
3. Assemble team	7
4. Collect prior water quality data	6
5. Obtain equipment	7
6. Identify streams	None
7. Decide on readings needed	4, 8
8. Review federal and state guidelines	None
9. Write report	11
10. Distribute equipment	3, 5
11. Analyze new data	2